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20 September 2026

Data 260 Homework 3

Personal Configuration and Domain

SID4	0932
PORT_BASE	8032
PREFIX	s0932
SEED	0932
VERIFY_SEED	260932
DOMAIN_ID	4 (open-source package vulnerabilities)

Hardware: nvidia geforce 4070, msi cpu

Local model used: ollama qwen3:8b

Tagged commit hash repo with most updated codes (submission on canvas only):

<https://github.com/n-k-leung/data260-0932>

Part 1: Fast API

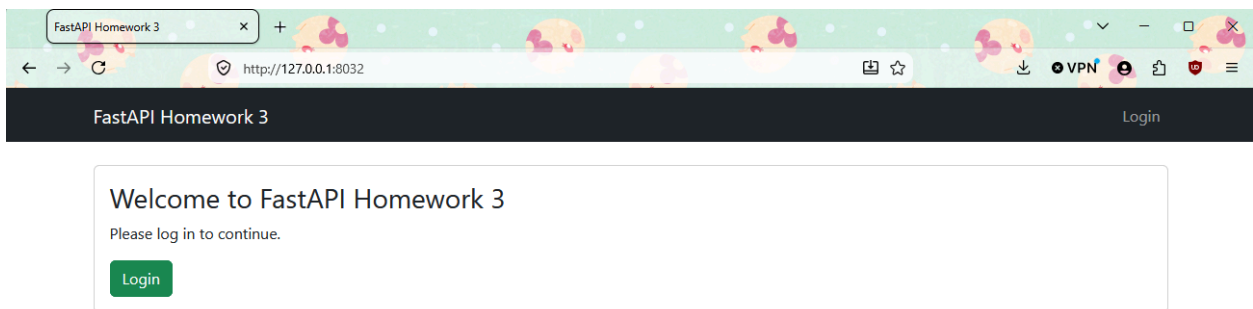
1. Routes - Handled Separately in a router:

- Home Page (/): Display a welcome message for your domain application. Show a link to the login page if the user is not logged in. Show a link to the dashboard and log out if the user is logged in.

```

21 @router.get("/")
22 def home(request: Request):
23     """
24     Home page route.
25
26     - Checks if a user is logged in using the session
27     - Passes user info to the template
28     """
29     user = request.session.get("user")
30
31     return templates.TemplateResponse(
32
33         request=request,
34         name="index.html",
35         context={
36             "user": user
37         }
38     )

```

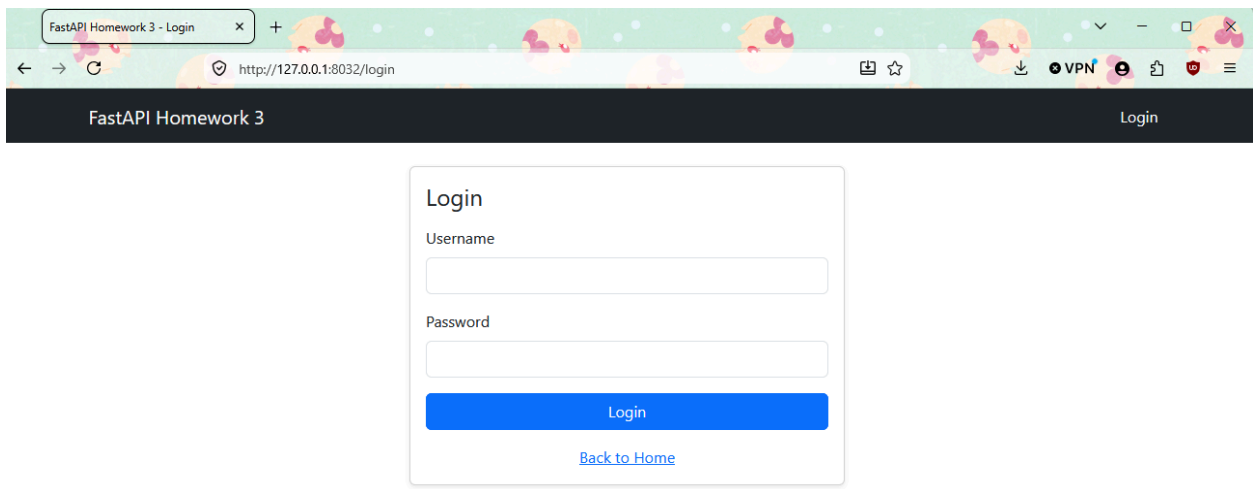


- Login Page (/login): Display a login form with fields for username and password. Validate the credentials and log the user in if they are correct. Redirect to the dashboard on successful login. Display a Bootstrap alert message on the login page when a user enters invalid credentials.

```

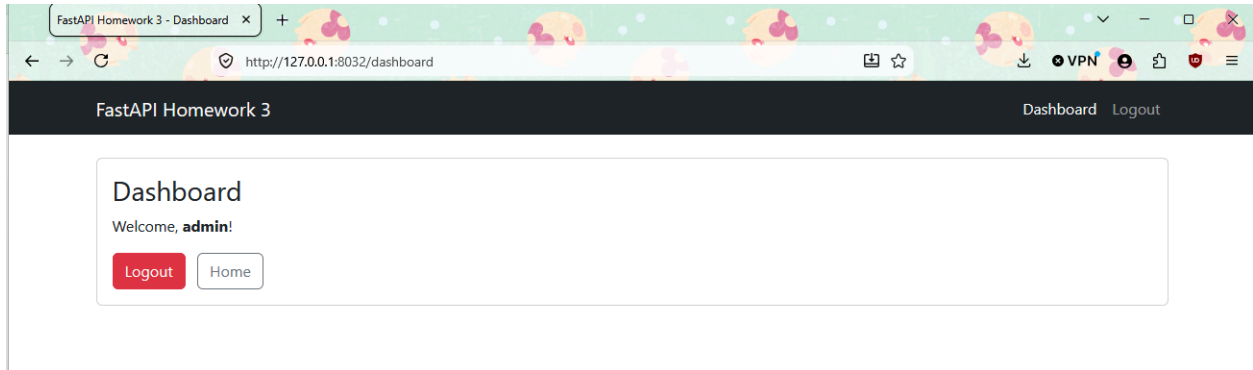
41 @router.get("/login")
42 def login_page(request: Request):
43     """
44     Displays the login form.
45
46     If the user is already logged in,
47     the template can choose what to display.
48     """
49     user = request.session.get("user")
50     #implementing error message
51     error = request.session.pop("login_error", None)
52
53     return templates.TemplateResponse(
54         request= request,
55         name="login.html",
56         context={
57             "user": user,
58             "error": error
59         }
60     )
61

```



The screenshot shows a web browser window with the title 'FastAPI Homework 3 - Login'. The address bar shows the URL 'http://127.0.0.1:8032/login'. The page content includes a dark header with 'FastAPI Homework 3' on the left and 'Login' on the right. The main content area features a white login form with the title 'Login'. Inside the form, there are two input fields: 'Username' and 'Password'. Below these fields is a blue 'Login' button. At the bottom of the form is a blue link labeled 'Back to Home'.

Valid login redirection



Invalid login

```
63 @router.post("/login")
64 def login(request: Request, username: str = Form(...), password: str = Form(...)):
65     """
66     Handles login form submission.
67
68     - Reads username and password from the form
69     - Validates credentials
70     - Stores user info in session if valid
71     """
72     if username == VALID_USERNAME and password == VALID_PASSWORD:
73         # Store logged-in user in session
74         request.session["user"] = username
75
76         # Redirect user to dashboard
77         return RedirectResponse(
78             url="/dashboard",
79             status_code=HTTP_302_FOUND
80         )
81
82     # If credentials are invalid:
83     # Redirect back to login page
84     #
85     # No error message is shown intentionally.
86     # Students are expected to add Bootstrap alerts.
87     request.session["login_error"] = "Invalid user or password"
88     return RedirectResponse(
89         url="/login",
90         status_code=HTTP_302_FOUND
91     )
92
93
```



Login

Username

test

Password

....

Login

[Back to Home](#)



Login

Danger! Invalid user or password. Please login again.

Username

Password

Login

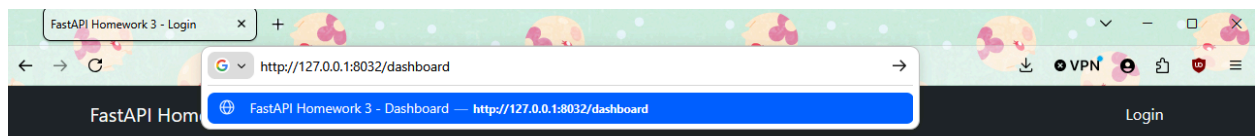
[Back to Home](#)

- Dashboard Page (/dashboard): Display a welcome message with the user's name. Show a logout link. Protect this route so only logged-in users can access it. Non logged in user trying to access dashboard

```

94 @router.get("/dashboard")
95 def dashboard(request: Request):
96     """
97     Protected route.
98
99     - Only accessible if user is logged in
100     - Redirects to login page if session is missing
101     """
102     user = request.session.get("user")
103
104     # If user is not logged in, block access
105     if not user:
106         return RedirectResponse(
107             url="/login",
108             status_code=HTTP_302_FOUND
109         )
110
111     # If user is logged in, render dashboard
112     return templates.TemplateResponse(
113         request=request,
114         name="dashboard.html",
115         context={
116             "user": user
117         }
118     )
119

```



Login

Username

Password

Login

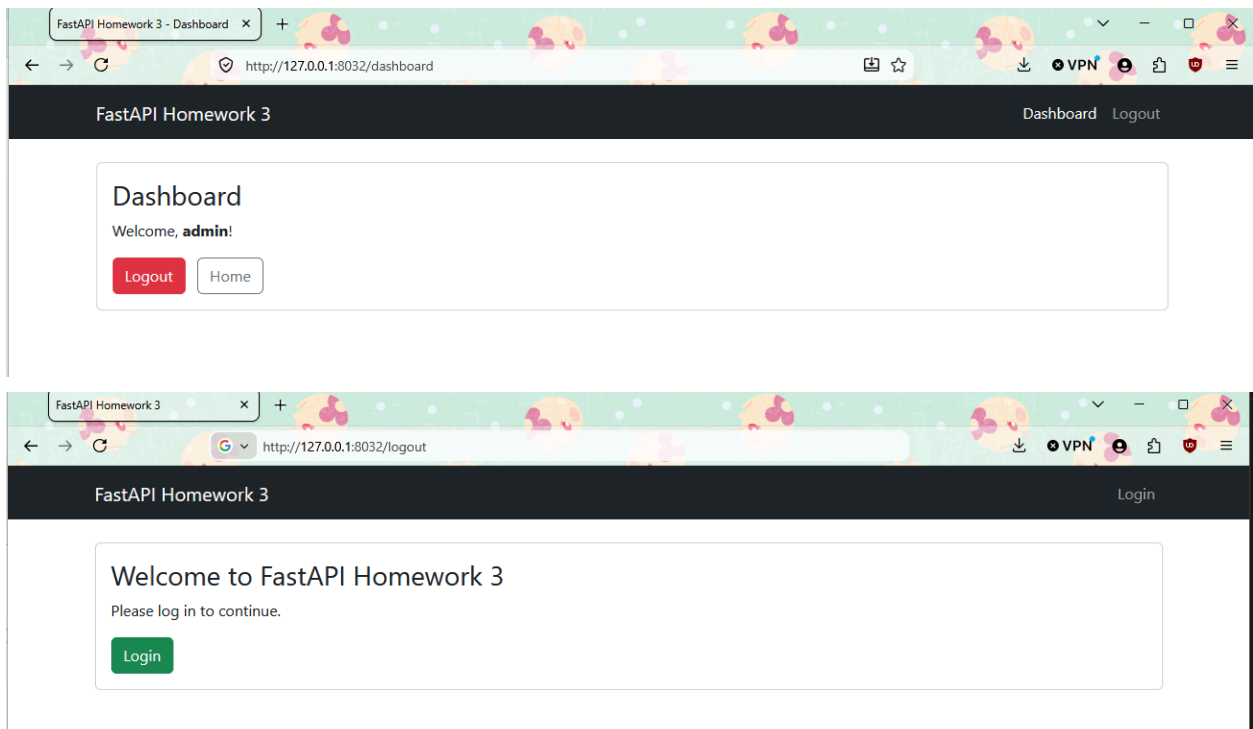
[Back to Home](#)

- Logout (/logout): Destroy the session and redirect the user to the home page.

```

121 @router.get("/logout")
122 def logout(request: Request):
123     """
124     Logs the user out.
125
126     - Clears all session data
127     - Redirects back to home page
128     """
129     request.session.clear()
130
131     return RedirectResponse(
132         url="/",
133         status_code=HTTP_302_FOUND
134     )
135

```



2. Session Management:

Use Starlette's SessionMiddleware to manage user sessions. Store the logged-in user's information in the session.

```

1  # Secret key for session signing
2  SECRET_KEY = os.getenv("SECRET_KEY", "dev-only-secret-key")
3
4  # Enable session support
5  app.add_middleware(
6      SessionMiddleware,
7      secret_key=SECRET_KEY,
8      https_only=True,
9      same_site="lax",
10     max_age=3600
11 )

```

Ensure that the session cookie is secure. Capture and include the Set-Cookie response header showing all three attributes.

Logged in valid user

The screenshot shows a web browser window with the address bar at `http://127.0.0.1:8032/dashboard`. The page title is "FastAPI Homework 3" and the URL is `http://127.0.0.1:8032/dashboard`. The dashboard content includes a "Welcome, admin!" message and two buttons: "Logout" (red) and "Home" (white). Below the dashboard, the Chrome DevTools Storage panel is open, showing the "Cookies" section. The "session" cookie is selected, and its details are displayed in the table below.

Name	Value	Domain	Path	Expires / Max-Age	Size	HttpOnly	Secure	SameSite	Last Accessed	Partition Key
session	eyJ1c2VyfjogImFkbWln0=arfm3QkQdPjmTyAh3eqUeTtZ...	127.0.0.1	/	Mon, 21 Sep 2026 18:18:2...	66	true	true	Lax	Mon, 21 Sep 2026 17:18:2...	

FastAPI Homework 3 - Dashboard

http://127.0.0.1:8032/dashboard

FastAPI Homework 3DashboardLogout

Dashboard

Welcome, admin!

LogoutHome

InspectorConsoleDebuggerNetworkStyle EditorPerformanceMemoryStorageAccessibilityApplication

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	POST	127.0.0.1:8032	login	document	html (...)	1.25 kB	0 B
200	GET	127.0.0.1:8032	dashboard	document	html	1.13 kB	981 B
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	33.19 ...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

HeadersCookiesRequestResponseTimings

POST http://127.0.0.1:8032/login

Status302 Found

VersionHTTP/1.1

Transferred1.25 kB (0 B size)

Referrer Policystrict-origin-when-cross-origin

Request PriorityHighest

DNS ResolutionSystem

Response Headers (266 B)Raw

content-length: 0

date: Mon, 21 Sep 2026 17:18:20 GMT

location: /dashboard

server: uvicorn

set-cookie: session=eyJ1c2VyJogImFkbWl0bn0=arFm3Q1QdPjmTyAh3eqUeTzbspp5-NQw; path=/; Max-Age=3600; httpOnly; sameSite=lax; secure

vary: Cookie

Request Headers (611 B)Raw

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8

Accept-Encoding: gzip, deflate, br, zstd

Accept-Language: en-US,en;q=0.9

Connection: keep-alive

Content-Length: 32

Content-Type: application/x-www-form-urlencoded

Host: 127.0.0.1:8032

Origin: http://127.0.0.1:8032

Priority: u=0,i

Referer: http://127.0.0.1:8032/login

Sec-Fetch-Dest: document

Sec-Fetch-Mode: navigate

Sec-Fetch-Site: same-origin

Sec-Fetch-User: ?1

Sec-GPC: 1

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:156.0) Gecko/20100101 Firefox/156.0

InspectorConsoleDebuggerNetworkStyle EditorPerformanceMemoryStorageAccessibilityApplication

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	POST	127.0.0.1:8032	login	document	html (...)	1.25 kB	0 B
200	GET	127.0.0.1:8032	dashboard	document	html	1.13 kB	981 B
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	33.19 ...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

HeadersCookiesRequestResponseTimings

Filter Cookies

Response Cookies

session:

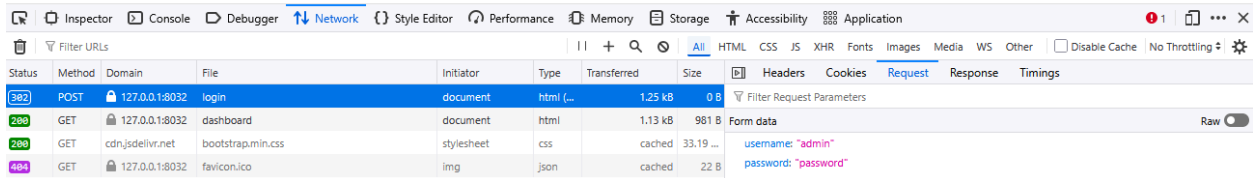
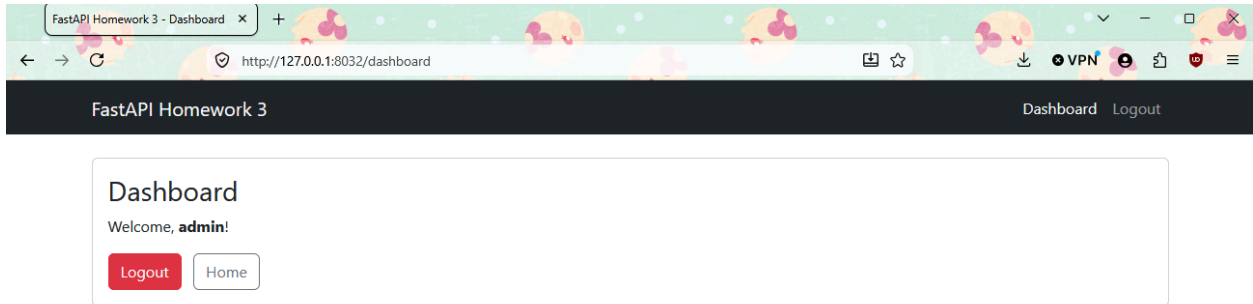
httpOnly: true

path: "/"

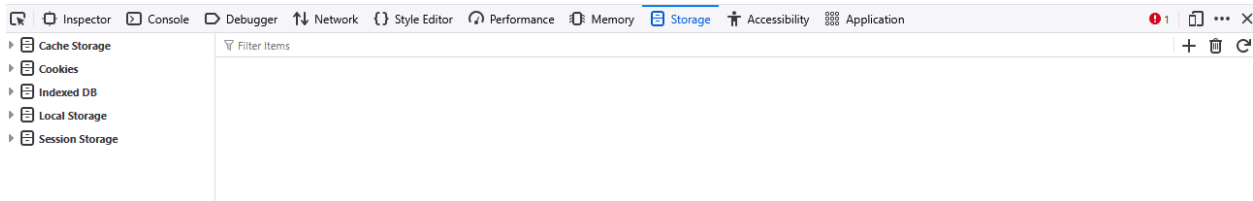
sameSite: "Lax"

secure: true

value: "eyJ1c2VyJogImFkbWl0bn0=arFm3Q1QdPjmTyAh3eqUeTzbspp5-NQw"



Logged out



FastAPI Homework 3

Welcome to FastAPI Homework 3

Please log in to continue.

Login

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	GET	127.0.0.1:8032	logout	document	html (...)	1.11 kB	0 B
200	GET	127.0.0.1:8032	/	document	html	1.03 kB	883 B
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	33.19 ...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

Headers Cookies Request Response Timings

GET http://127.0.0.1:8032/logout

Status: 302 Found

Version: HTTP/1.1

Transferred: 1.11 kB (0 B size)

Referrer Policy: strict-origin-when-cross-origin

Request Priority: Highest

DNS Resolution: System

Response Headers (227 B)

content-length: 0

date: Mon, 21 Sep 2026 17:24:49 GMT

location: /

server: uvicorn

set-cookie: session=null; path=/; expires=Thu, 01 Jan 1970 00:00:00 GMT; httponly; samesite=lax; secure

vary: Cookie

Request Headers (592 B)

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8

Accept-Encoding: gzip, deflate, br, zstd

Accept-Language: en-US,en;q=0.9

Connection: keep-alive

Cookie: session=eyJ1c2VyJmFkbWln0=arFm3Q,tQdPjmTyAh3eqUeTzbsppS-NQw

Host: 127.0.0.1:8032

Priority: u=0, l

Referer: http://127.0.0.1:8032/dashboard

Sec-Fetch-Dest: document

Sec-Fetch-Mode: navigate

Sec-Fetch-Site: same-origin

Sec-Fetch-User: ?1

Sec-GPC: 1

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:156.0) Gecko/20100101 Firefox/156.0

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	GET	127.0.0.1:8032	logout	document	html (...)	1.11 kB	0 B
200	GET	127.0.0.1:8032	/	document	html	1.03 kB	883 B
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	33.19 ...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

Headers Cookies Request Response Timings

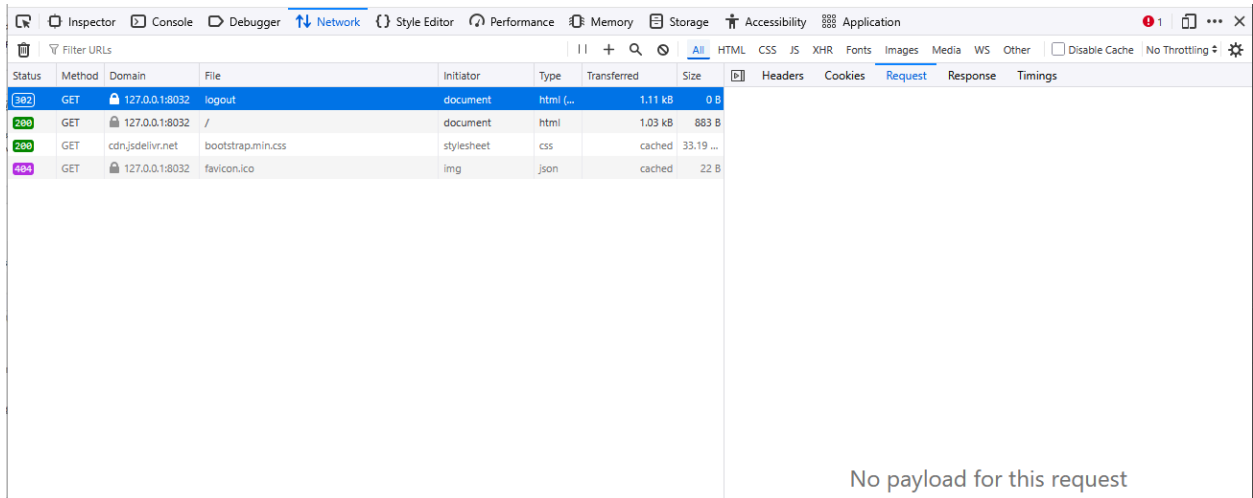
Filter Cookies

Response Cookies

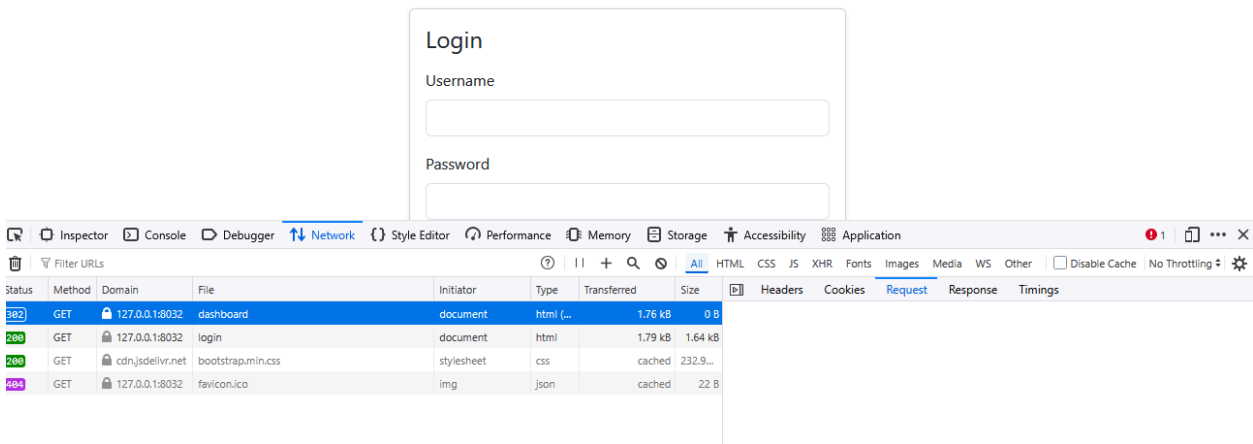
session: expires: "1970-01-01T00:00:00.000Z" httpOnly: true path: "/" samesite: "Lax" secure: true value: "null"

Request Cookies

session: "eyJ1c2VyJmFkbWln0=arFm3Q,tQdPjmTyAh3eqUeTzbsppS-NQw"



Trying to access dashboard with no user logged in



FastAPI Homework 3 - Login x +

http://127.0.0.1:8032/login

FastAPI Homework 3 Login

Login

Username

Password

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	GET	127.0.0.1:8032	dashboard	document	html (...)	1.76 kB	0 B
200	GET	127.0.0.1:8032	login	document	html	1.79 kB	1.64 kB
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	232.9...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

Filter Headers

GET http://127.0.0.1:8032/dashboard

Status: 302 Found

Version: HTTP/1.1

Transferred: 1.76 kB (0 B size)

Request Priority: Highest

DNS Resolution: System

Response Headers (127 B)

content-length: 0

date: Mon, 21 Sep 2026 17:26:38 GMT

location: /login

server: uvicorn

vary: Cookie

Request Headers (469 B)

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8

Accept-Encoding: gzip, deflate, br, zstd

Accept-Language: en-US,en;q=0.9

Connection: keep-alive

Host: 127.0.0.1:8032

Priority: u=0, i

Sec-Fetch-Dest: document

Sec-Fetch-Mode: navigate

Sec-Fetch-Site: none

Sec-Fetch-User: ?1

Sec-GPC: 1

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:156.0) Gecko/20100101 Firefox/156.0

FastAPI Homework 3 - Login x +

http://127.0.0.1:8032/login

FastAPI Homework 3 Login

Login

Username

Password

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	GET	127.0.0.1:8032	dashboard	document	html (...)	1.76 kB	0 B
200	GET	127.0.0.1:8032	login	document	html	1.79 kB	1.64 kB
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	232.9...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

Filter Headers

GET http://127.0.0.1:8032/dashboard

Status: 302 Found

Version: HTTP/1.1

Transferred: 1.76 kB (0 B size)

Request Priority: Highest

DNS Resolution: System

Response Headers (127 B)

content-length: 0

date: Mon, 21 Sep 2026 17:26:38 GMT

location: /login

server: uvicorn

vary: Cookie

Request Headers (469 B)

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8

Accept-Encoding: gzip, deflate, br, zstd

Accept-Language: en-US,en;q=0.9

Connection: keep-alive

Host: 127.0.0.1:8032

Priority: u=0, i

Sec-Fetch-Dest: document

Sec-Fetch-Mode: navigate

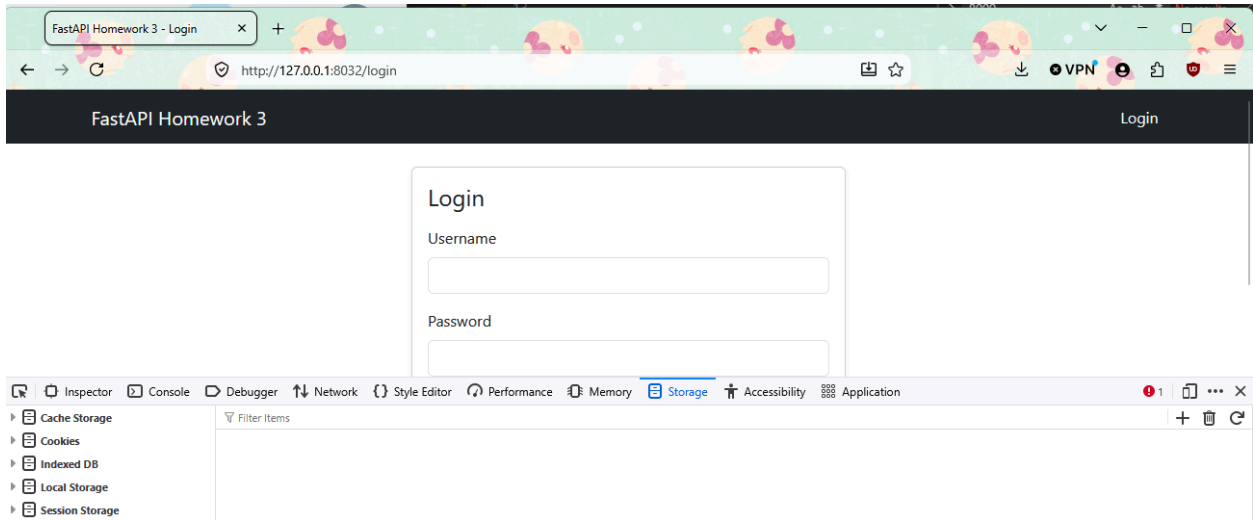
Sec-Fetch-Site: none

Sec-Fetch-User: ?1

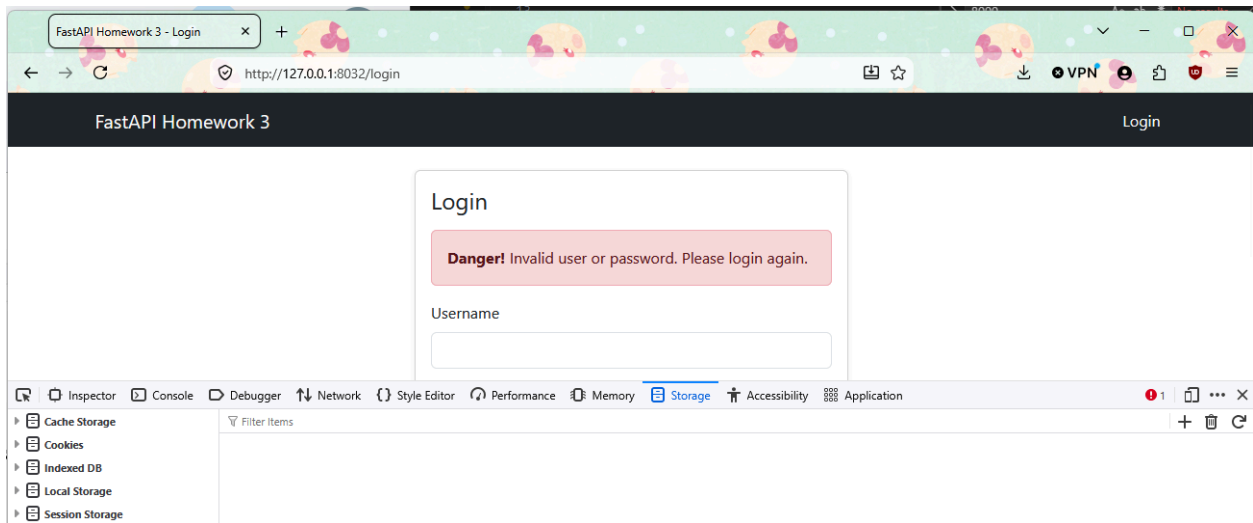
Sec-GPC: 1

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:156.0) Gecko/20100101 Firefox/156.0



Invalid user login



FastAPI Homework 3 - Login

http://127.0.0.1:8032/login

FastAPI Homework 3

Login

Danger! Invalid user or password. Please login again.

Username

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	POST	127.0.0.1:8032	login	document	html (...)	2.09 kB	0 B
200	GET	127.0.0.1:8032	login	document	html	2.05 kB	1.80 kB
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	232.9...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

Headers Cookies Request Response Timings

Filter Headers

Block Resend

POST http://127.0.0.1:8032/login

Status: 302 Found

Version: HTTP/1.1

Transferred: 2.09 kB (0 B size)

Referrer Policy: strict-origin-when-cross-origin

Request Priority: Highest

DNS Resolution: System

Response Headers (298 B)

content-length: 0

date: Mon, 21 Sep 2026 17:28:17 GMT

location: /login

server: uvicorn

set-cookie: session=eyJsb2dpbi9icmVjci6iCjBnZhbGkiHVzZXlgb3IgcGFzc3dvcmlQIjQ=..arFpMQ.P291O_WaAXsZ_-ExzyKIIVQbNU; path=/; Max-Age=3600; httponly; samesite=lax; secure

vary: Cookie

Request Headers (611 B)

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8

Accept-Encoding: gzip, deflate, br, zstd

Accept-Language: en-US,en;q=0.9

Connection: keep-alive

Content-Length: 27

Content-Type: application/x-www-form-urlencoded

Host: 127.0.0.1:8032

Origin: http://127.0.0.1:8032

Priority: u=0,i

Referer: http://127.0.0.1:8032/login

Sec-Fetch-Dest: document

Sec-Fetch-Mode: navigate

Sec-Fetch-Site: same-origin

Sec-Fetch-User: ?1

Sec-GPC: 1

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:156.0) Gecko/20100101 Firefox/156.0

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	POST	127.0.0.1:8032	login	document	html (...)	2.09 kB	0 B
200	GET	127.0.0.1:8032	login	document	html	2.05 kB	1.80 kB
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	232.9...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

Headers Cookies Request Response Timings

Filter Cookies

Response Cookies

session: httpOnly: true, path: /, samesite: "Lax", secure: true, value: "eyJsb2dpbi9icmVjci6iCjBnZhbGkiHVzZXlgb3IgcGFzc3dvcmlQIjQ=..arFpMQ.P291O_WaAXsZ_-ExzyKIIVQbNU"

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	POST	127.0.0.1:8032	login	document	html (...)	2.09 kB	0 B
200	GET	127.0.0.1:8032	login	document	html	2.05 kB	1.80 kB
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	232.9...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

Headers Cookies Request Response Timings

Filter Request Parameters

Form data

username: "test", password: "test"

Implement an idle session timeout. Prove that an expired or logged-out session cannot be reused to reach /dashboard.

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw03> python main.py
INFO: Will watch for changes in these directories: ['C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw03']
INFO: Uvicorn running on http://127.0.0.1:8032 (Press CTRL+C to quit)
INFO: Started reloader process [17812] using StatReload
INFO: Started server process [18420]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:63583 - "GET /dashboard HTTP/1.1" 302 Found
INFO: 127.0.0.1:63583 - "GET /login HTTP/1.1" 200 OK
INFO: 127.0.0.1:63583 - "GET /login HTTP/1.1" 200 OK
INFO: 127.0.0.1:63583 - "GET /login HTTP/1.1" 200 OK
INFO: 127.0.0.1:63598 - "POST /login HTTP/1.1" 302 Found
INFO: 127.0.0.1:63598 - "GET /dashboard HTTP/1.1" 200 OK
INFO: 127.0.0.1:63611 - "GET /dashboard HTTP/1.1" 302 Found
INFO: 127.0.0.1:63611 - "GET /login HTTP/1.1" 200 OK
INFO: 127.0.0.1:63615 - "GET /dashboard HTTP/1.1" 302 Found
INFO: 127.0.0.1:63615 - "GET /login HTTP/1.1" 200 OK
```

```
21 def idle(request:Request) -> bool:
22     last_activity = request.session.get("last_activity")
23     #user not logged in yet so always false
24     if last_activity is None:
25         return False
26     seconds_count = time.time() - last_activity
27     return seconds_count > 60
```

```

72 @router.post("/login")
73 ✓ def login(request: Request, username: str = Form(...), password: str = Form(...)):
74 ✓     """
75     Handles login form submission.
76
77     - Reads username and password from the form
78     - Validates credentials
79     - Stores user info in session if valid
80     """
81 ✓     if username == VALID_USERNAME and password == VALID_PASSWORD:
82         # Store logged-in user in session
83         request.session["user"] = username
84         #start counting how long user has been idle
85         request.session["last_activity"] = time.time()
86
87         # Redirect user to dashboard
88 ✓         return RedirectResponse(
89             url="/dashboard",
90             status_code=HTTP_302_FOUND
91         )
92
93     # If credentials are invalid:
94     # Redirect back to login page
95     #
96     # No error message is shown intentionally.
97     # Students are expected to add Bootstrap alerts.
98     request.session["login_error"] = "Invalid user or password"
99 ✓     return RedirectResponse(
100         url="/login",
101         status_code=HTTP_302_FOUND
102     )

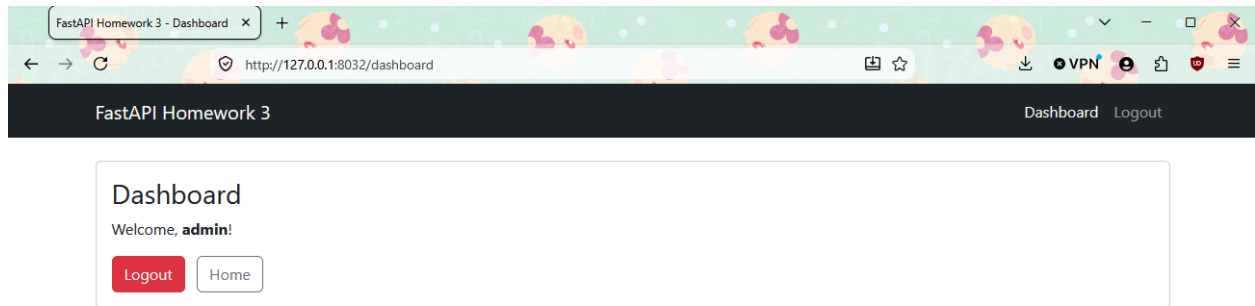
```

```

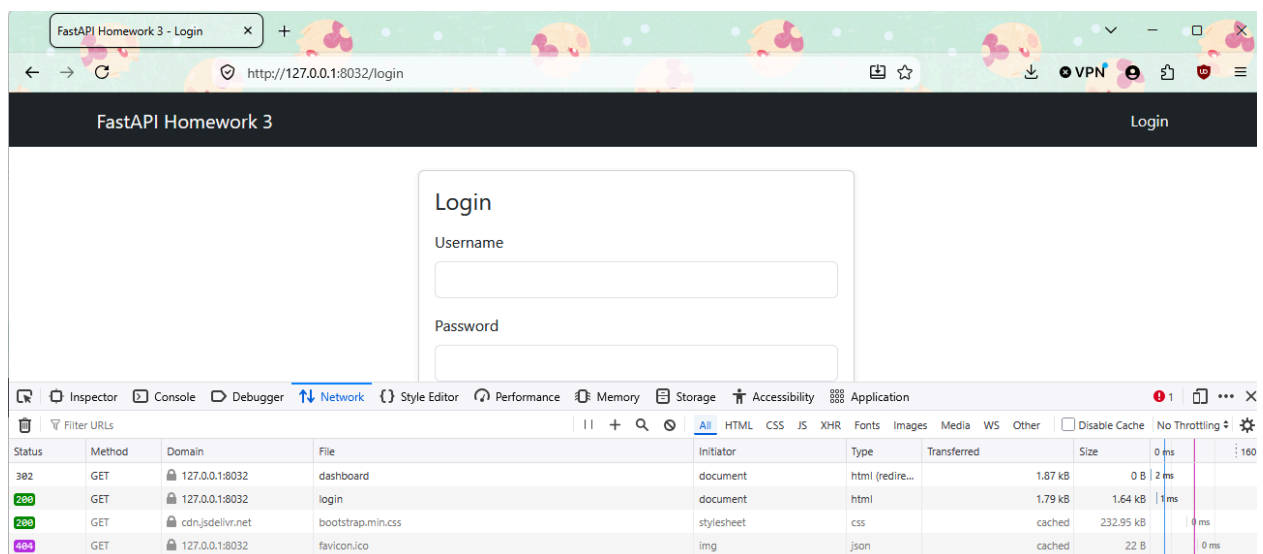
102 @router.get("/dashboard")
103 def dashboard(request: Request):
104     """
105     Protected route.
106
107     - Only accessible if user is logged in
108     - Redirects to login page if session is missing
109     """
110     user = request.session.get("user")
111
112     # If user is not logged in, block access
113     if not user:
114         return RedirectResponse(
115             url="/login",
116             status_code=HTTP_302_FOUND
117         )
118
119     #time user out and redirect to login
120     if idle(request):
121         request.session.clear()
122         return RedirectResponse(
123             url="/login",
124             status_code=HTTP_302_FOUND
125         )
126     #update how long user has been idle if active within the 60 sec count
127     request.session["last_activity"] = time.time()
128     # If user is logged in, render dashboard
129     return templates.TemplateResponse(
130         request=request,
131         name= "dashboard.html",
132         context={
133             "user": user
134         }
135     )
136

```

On dashboard and then timeout and redirect to login page



Status	Method	Domain	File	Initiator	Type	Transferred	Size	0 ms	64
302	POST	127.0.0.1:8032	login	document	html (redire...		1.30 kB	0 B	2 ms
200	GET	127.0.0.1:8032	dashboard	document	html		1.31 kB	981 B	2 ms
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css		cached	33.19 kB	0 ms
404	GET	127.0.0.1:8032	favicon.ico	img	json		cached	22 B	0 ms



Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	GET	127.0.0.1:8032	dashboard	document	html (...)	1.87 kB	0 B
200	GET	127.0.0.1:8032	login	document	html	1.79 kB	1.64 kB
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	232.9...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

Headers Cookies Request Response Timings

Filter Headers

GET http://127.0.0.1:8032/dashboard

Status: 302 Found

Version: HTTP/1.1

Transferred: 1.87 kB (0 B size)

Referrer Policy: strict-origin-when-cross-origin

Request Priority: Highest

DNS Resolution: System

Response Headers (232 B)

content-length: 0

date: Mon, 21 Sep 2026 17:48:12 GMT

location: /login

server: uvicorn

set-cookie: session=null; path=/; expires=Thu, 01 Jan 1970 00:00:00 GMT; httponly; samesite=lax; secure

vary: Cookie

Request Headers (639 B)

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8

Accept-Encoding: gzip, deflate, br, zstd

Accept-Language: en-US,en;q=0.9

Connection: keep-alive

Cookie: session=eyJ1c2VyJogImFkbWl1wglmXhc3RlYWN0aXZpdHkiOiA4NzkxMDYyODUzLjYyMjA5NjZ9.arFttQw1lcEtdZmEq13f7N1H1YX5YpA4

Host: 127.0.0.1:8032

Priority: u=0,i

Referer: http://127.0.0.1:8032/login

Sec-Fetch-Dest: document

Sec-Fetch-Mode: navigate

Sec-Fetch-Site: same-origin

Sec-Fetch-User: ?1

Sec-GPC: 1

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:156.0) Gecko/20100101 Firefox/156.0

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	GET	127.0.0.1:8032	dashboard	document	html (...)	1.87 kB	0 B
200	GET	127.0.0.1:8032	login	document	html	1.79 kB	1.64 kB
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	232.9...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

Headers Cookies Request Response Timings

Filter Cookies

Response Cookies

session: expires="1970-01-01T00:00:00.000Z" httpOnly: true path: "/" samesite: "Lax" secure: true value: "null"

Request Cookies

session: "eyJ1c2VyJogImFkbWl1wglmXhc3RlYWN0aXZpdHkiOiA4NzkxMDYyODUzLjYyMjA5NjZ9.arFttQw1lcEtdZmEq13f7N1H1YX5YpA4"

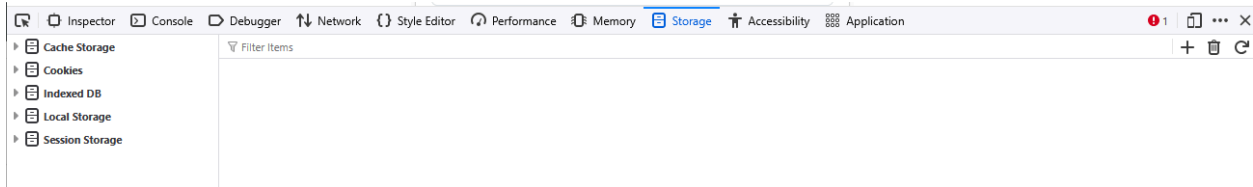
Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

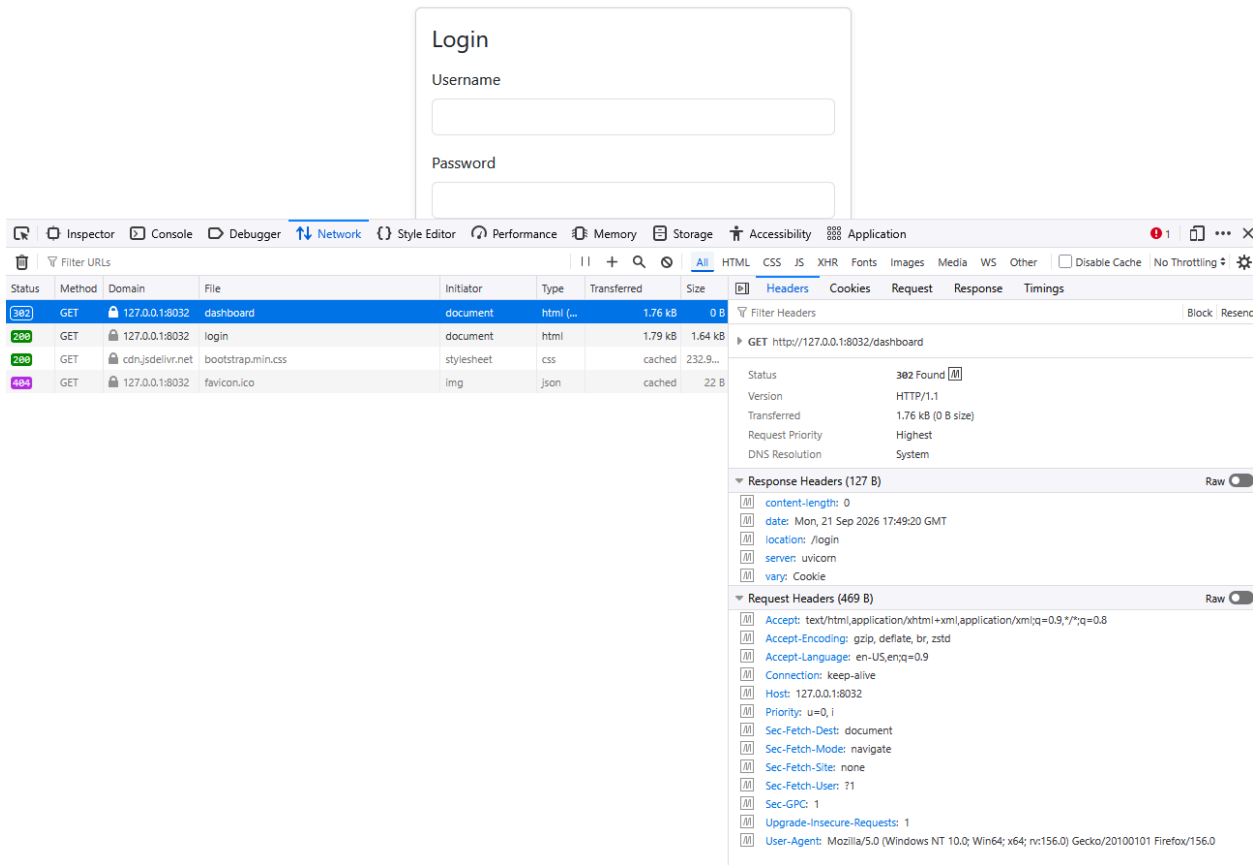
Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	GET	127.0.0.1:8032	dashboard	document	html (...)	1.87 kB	0 B
200	GET	127.0.0.1:8032	login	document	html	1.79 kB	1.64 kB
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	232.9...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

Headers Cookies Request Response Timings

No payload for this request



Cannot go back to dashboard



Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	GET	127.0.0.1:8032	dashboard	document	html (...)	1.76 kB	0 B
200	GET	127.0.0.1:8032	login	document	html	1.79 kB	1.64 kB
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	232.9...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

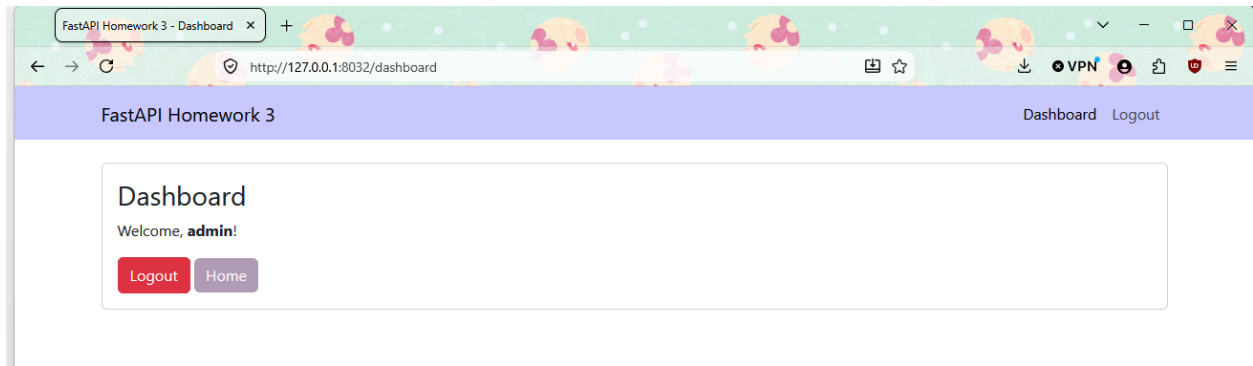
No cookies for this request

Status	Method	Domain	File	Initiator	Type	Transferred	Size
302	GET	127.0.0.1:8032	dashboard	document	html (...)	1.76 kB	0 B
200	GET	127.0.0.1:8032	login	document	html	1.79 kB	1.64 kB
200	GET	cdn.jsdelivr.net	bootstrap.min.css	stylesheet	css	cached	232.9...
404	GET	127.0.0.1:8032	favicon.ico	img	json	cached	22 B

No payload for this request

Storage Area	Filter Items
Cache Storage	
Cookies	
Indexed DB	
Local Storage	
Session Storage	

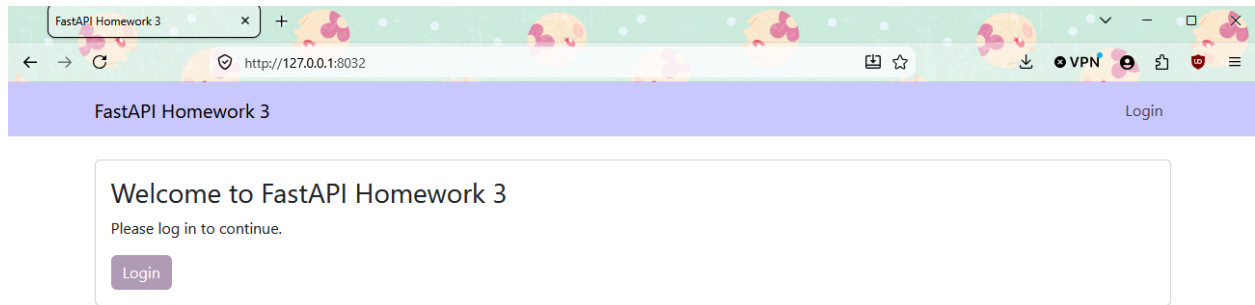
3. Styling with Bootstrap: Explore and use Bootstrap to style all pages. Make the application responsive and visually appealing. Use Bootstrap components, such as the Navbar for navigation, Cards for forms and content, Buttons for actions, and Alerts for messages. Include one screenshot per route page.
Dashboard.html



```
data260-0932 > reports > hw03 > templates > <> dashboard.html
```

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-scale=1" />
6   <title>FastAPI Homework 3 - Dashboard</title>
7   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
8 </style>
9   .btn-purple{
10     background-color:#B39EB5;
11     color: white;
12     border:none;
13   }
14   .btn-purple:hover{
15     background-color:#805B87;
16     color: white;
17     border:none;
18   }
19 </style>
20 </head>
21 <body>
22
23 <nav class="navbar navbar-expand-lg navbar-light" style="background-color:#CCCCFF;">
24   <div class="container">
25     <a class="navbar-brand" href="/">FastAPI Homework 3</a>
26     <div class="navbar-nav">
27       <a class="nav-link active" href="/dashboard">Dashboard</a>
28       <a class="nav-link" href="/logout">Logout</a>
29     </div>
30   </div>
31 </nav>
32
33 <div class="container mt-4">
34   <div class="card">
35     <div class="card-body">
36       <h3 class="card-title">Dashboard</h3>
37       <p class="card-text">Welcome, <strong>{{ user }}</strong>!</p>
38
39       <a class="btn btn-danger" href="/logout">Logout</a>
40       <a class="btn btn-purple" href="/">Home</a>
41     </div>
42   </div>
43 </div>
44
45 </body>
46 </html>
47
```

Index.html



```

data260-0932 > reports > hw03 > templates > <> index.html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8" />
5      <meta name="viewport" content="width=device-width, initial-scale=1" />
6      <title>FastAPI Homework 3</title>
7      <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
8      <style>
9          .btn-purple{
10             background-color:#839EB5;
11             color: white;
12             border:none;
13         }
14         .btn-purple:hover{
15             background-color:#805B87;
16             color: white;
17             border:none;
18         }
19     </style>
20 </head>
21 <body>
22
23 <nav class="navbar navbar-expand-lg navbar-light" style="background-color:#CCCCFF;">
24     <div class="container">
25         <a class="navbar-brand" href="/">FastAPI Homework 3</a>
26         <div class="navbar-nav">
27             {% if user %}
28                 <a class="nav-link" href="/dashboard">Dashboard</a>
29                 <a class="nav-link" href="/logout">Logout</a>
30             {% else %}
31                 <a class="nav-link" href="/login">Login</a>
32             {% endif %}
33         </div>
34     </div>
35 </nav>
36
37 <div class="container mt-4">
38     <div class="card">
39         <div class="card-body">
40             <h3 class="card-title">Welcome to FastAPI Homework 3</h3>
41
42             {% if user %}
43                 <p class="card-text">You are logged in as <strong>{{ user }}</strong>.</p>
44                 <a class="btn btn-purple" href="/dashboard">Go to Dashboard</a>
45                 <a class="btn btn-outline-danger ms-2" href="/logout">Logout</a>
46             {% else %}
47                 <p class="card-text">Please log in to continue.</p>
48                 <a class="btn btn-purple" href="/login">Login</a>
49             {% endif %}
50         </div>
51     </div>
52 </div>
53
54 </body>
55 </html>
56

```

Login.html

FastAPI Homework 3 - Login

+

⌵

—

□

✕

←

→

↻

🔒

http://127.0.0.1:8032/login

📄

☆

⬇

VPN

🔒

📄

🔒

☰

FastAPI Homework 3

Login

Login

Username

admin

Password

••••••••

Login

Back to Home

data260-0932 > reports > hw03 > templates > <> login.html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8" />
5      <meta name="viewport" content="width=device-width, initial-scale=1" />
6      <title>FastAPI Homework 3 - Login</title>
7      <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
8  </head>
9  <body>
10
11  <nav class="navbar navbar-expand-lg navbar-light" style="background-color:#CCCCFF;">
12      <div class="container">
13          <a class="navbar-brand" href="/">FastAPI Homework 3</a>
14          <div class="navbar-nav">
15              {% if user %}
16                  <a class="nav-link" href="/dashboard">Dashboard</a>
17                  <a class="nav-link" href="/logout">Logout</a>
18              {% else %}
19                  <a class="nav-link active" href="/login">Login</a>
20              {% endif %}
21          </div>
22      </div>
23  </nav>
24
25  <div class="container mt-4">
26      <div class="row justify-content-center">
27          <div class="col-md-5">
28              <div class="card shadow-sm">
29                  <div class="card-body">
30                      <h4 class="card-title mb-3">Login</h4>
31
32                      <!-- Intentionally no alert shown for invalid login.
33                      | Students can add a Bootstrap Alert as an extra step. -->
34                      {% if error %}
35                          <div class="alert alert-danger">
36                              <strong>Danger!</strong> Invalid user or password. Please login again.
37                          </div>
38                      {% endif %}
39                      <form method="post" action="/login">
40                          <div class="mb-3">
41                              <label class="form-label">Username</label>
42                              <input class="form-control" type="text" name="username" required>
43                          </div>
44
45                          <div class="mb-3">
46                              <label class="form-label">Password</label>
47                              <input class="form-control" type="password" name="password" required>
48                          </div>
49
50                          <button class="btn btn-purple w-100" type="submit">Login</button>
51                      </form>
52
53                      <div class="text-center mt-3">
54                          <a href="/" class="btn btn-purple"> Back to Home</a>
55                      </div>
56                      <style>
57                          .btn-purple{
58                              background-color:#B39EB5;
59                              color: white;
60                              border:none;
61                          }
```

```
62  ✓      .btn-purple:hover{
63          background-color:#805B87;
64          color: white;
65          border:none;
66      }
67      </style>
68  </div>
69  </div>
70  </div>
71  </div>
72  </div>
73
74  </body>
75  </html>
76
```

Include only the auth.py code at the bottom and one screenshot of your templates directory.

```

data260-0932 > reports > hw03 > routers > auth.py > home
1  from fastapi import APIRouter, Request, Form
2  from fastapi.responses import RedirectResponse
3  from fastapi.templating import Jinja2Templates
4  from starlette.status import HTTP_302_FOUND
5  import time
6
7
8  # Create a router object
9  # This behaves like a mini FastAPI app
10 router = APIRouter()
11
12 # Configure Jinja2 templates directory
13 templates = Jinja2Templates(directory="templates")
14
15
16 # Hardcoded credentials for demo purposes only
17 # In real applications, credentials come from a database
18 VALID_USERNAME = "admin"
19 VALID_PASSWORD = "password"
20
21 def idle(request: Request) -> bool:
22     last_activity = request.session.get("last_activity")
23     #user not logged in yet so always false
24     if last_activity is None:
25         return False
26     seconds_count = time.time() - last_activity
27     return seconds_count > 60
28
29 @router.get("/")
30 def home(request: Request):
31     """
32     Home page route.
33
34     - Checks if a user is logged in using the session
35     - Passes user info to the template
36     """
37     user = request.session.get("user")
38
39     return templates.TemplateResponse(
40
41         request=request,
42         name="index.html",
43         context={
44             "user": user
45         }
46     )
47
48
49 @router.get("/login")
50 def login_page(request: Request):
51     """
52     Displays the login form.
53
54     If the user is already logged in,
55     the template can choose what to display.
56     """
57     user = request.session.get("user")
58
59     #implementing error message
60     error = request.session.pop("login_error", None)
61

```

```

62     return templates.TemplateResponse(
63         request=request,
64         name="login.html",
65         context={
66             "user": user,
67             "error": error
68         }
69     )
70
71
72 @router.post("/login")
73 def login(request: Request, username: str = Form(...), password: str = Form(...)):
74     """
75     Handles login form submission.
76
77     - Reads username and password from the form
78     - Validates credentials
79     - Stores user info in session if valid
80     """
81     if username == VALID_USERNAME and password == VALID_PASSWORD:
82         # Store logged-in user in session
83         request.session["user"] = username
84         # Start counting how long user has been idle
85         request.session["last_activity"] = time.time()
86
87         # Redirect user to dashboard
88         return RedirectResponse(
89             url="/dashboard",
90             status_code=HTTP_302_FOUND
91         )
92
93     # If credentials are invalid:
94     # Redirect back to login page
95     #
96     # No error message is shown intentionally.
97     # Students are expected to add Bootstrap alerts.
98     request.session["login_error"] = "Invalid user or password"
99     return RedirectResponse(
100         url="/login",
101         status_code=HTTP_302_FOUND
102     )
103
104
105 @router.get("/dashboard")
106 def dashboard(request: Request):
107     """
108     Protected route.
109
110     - Only accessible if user is logged in
111     - Redirects to login page if session is missing
112     """
113     user = request.session.get("user")
114
115     # If user is not logged in, block access
116     if not user:
117         return RedirectResponse(
118             url="/login",
119             status_code=HTTP_302_FOUND

```

```

105 @router.get("/dashboard")
106 def dashboard(request: Request):
107     """
108     Protected route.
109
110     - Only accessible if user is logged in
111     - Redirects to login page if session is missing
112     """
113     user = request.session.get("user")
114
115     # If user is not logged in, block access
116     if not user:
117         return RedirectResponse(
118             url="/login",
119             status_code=HTTP_302_FOUND
120         )
121
122     #time user out and redirect to login
123     if idle(request):
124         request.session.clear()
125         return RedirectResponse(
126             url="/login",
127             status_code=HTTP_302_FOUND
128         )
129     #update how long user has been idle if active within the 60 sec count
130     request.session["last_activity"] = time.time()
131     # If user is logged in, render dashboard
132     return templates.TemplateResponse(
133         request=request,
134         name="dashboard.html",
135         context={
136             "user": user
137         }
138     )
139
140
141 @router.get("/logout")
142 def logout(request: Request):
143     """
144     Logs the user out.
145
146     - Clears all session data
147     - Redirects back to home page
148     """
149     request.session.clear()
150
151     return RedirectResponse(
152         url="/",
153         status_code=HTTP_302_FOUND
154     )

```

```

▼ templates
  <> dashboard.html
  <> index.html
  <> login.html

```

Part 2: Compare Three LlamaIndex Chunking Techniques (Retrieval-Only RAG)

Dataset

Source of open source packages vulnerability: PyPI JSON API, used 12 python packages to meet byte requirement

```

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw03> python .\get_packages.py
Flask.txt, bytes: 6649, number of vuls: 8
numpy.txt, bytes: 376, number of vuls: 1
cryptography.txt, bytes: 29787, number of vuls: 24
PyYAML.txt, bytes: 2605, number of vuls: 4
Jinja2.txt, bytes: 11660, number of vuls: 18
Django.txt, bytes: 11830, number of vuls: 18
lxml.txt, bytes: 5813, number of vuls: 8
pillow.txt, bytes: 95553, number of vuls: 52
requests.txt, bytes: 10757, number of vuls: 12
urllib3.txt, bytes: 28924, number of vuls: 24
Twisted.txt, bytes: 27438, number of vuls: 24
paramiko.txt, bytes: 1282, number of vuls: 4
SOURCES.md created
CORPUS_MANIFEST.json created

```

data260-0932 > reports > hw03 > SOURCES.md

	Package	Version	Source URL	Accessed
1	Flask	0.12	https://pypi.org/pypi/flask/0.12/json	2026-09-20
2	numpy	1.21.0	https://pypi.org/pypi/numpy/1.21.0/json	2026-09-20
3	cryptography	3.2	https://pypi.org/pypi/cryptography/3.2/json	2026-09-20
4	PyYAML	5.3	https://pypi.org/pypi/pyyaml/5.3/json	2026-09-20
5	Jinja2	2.4.1	https://pypi.org/pypi/jinja2/2.4.1/json	2026-09-20
6	Django	1.11.28	https://pypi.org/pypi/django/1.11.28/json	2026-09-20
7	lxml	4.6.2	https://pypi.org/pypi/lxml/4.6.2/json	2026-09-20
8	pillow	8.2.0	https://pypi.org/pypi/pillow/8.2.0/json	2026-09-20
9	requests	2.3.0	https://pypi.org/pypi/requests/2.3.0/json	2026-09-20
10	urllib3	1.24.1	https://pypi.org/pypi/urllib3/1.24.1/json	2026-09-20
11	Twisted	18.4.0	https://pypi.org/pypi/twisted/18.4.0/json	2026-09-20
12	paramiko	2.4.1	https://pypi.org/pypi/paramiko/2.4.1/json	2026-09-20
13				
14				
15				

```

C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw03> cat SOURCES.md
1 dataset/twisted.txt, "byte_size": 27438, "sha256": "66a0ff3322cf6acdb976d0f5d09238e472ef643f5385720a05a1444", ("filename": "dataset/paramiko.txt", "byte_size": 1282, "sha256": "6a702b3f8a44078f781e5cc3ca21b7cf590f4b400949d79f61794258")

```

dataset

- cryptography.txt
- Django.txt
- Flask.txt
- Jinja2.txt
- lxml.txt
- numpy.txt
- paramiko.txt
- pillow.txt
- PyYAML.txt
- requests.txt
- Twisted.txt
- urllib3.txt

```

data260-0932 > reports > hw03 > get_packages.py > main
1  import json
2  import urllib.request
3  import hashlib
4  from datetime import date
5
6  packages=[
7      "flask/0.12",
8      "numpy/1.21.0",
9      "cryptography/3.2",
10     "pyyaml/5.3",
11     "jinja2/2.4.1",
12     "django/1.11.28",
13     "lxml/4.6.2",
14     "pillow/8.2.0",
15     "requests/2.3.0",
16     "urllib3/1.24.1",
17     "twisted/18.4.0",
18     "paramiko/2.4.1",
19 ]
20
21 def get_version(version):
22     url=f"https://pypi.org/pypi/{version}/json"
23     with urllib.request.urlopen(url) as r:
24         return json.load(r)
25 #find the sha256 encoding
26 def get_sha(path):
27     with open(path, "rb") as f:
28         files_byte=f.read()
29     return hashlib.sha256(files_byte).hexdigest()
30
31 def main():
32     sources=[]
33     manifest=[]
34     access_date=date.today().isoformat()
35     for p in packages:
36         data=get_version(p)
37         name = data["info"]["name"]
38         vuls = data.get("vulnerabilities", [])
39         lines= [f"package: {name} (version:{p})", f"vulnerability count: {len(vuls)}", ""]
40         for v in vuls:
41             lines.append(f"id: {v.get('id')}")
42             lines.append(f"aliases: {', '.join(v.get('aliases', []))}")
43             lines.append(f"summary: {v.get('summary', '')}")
44             lines.append(f"details: {v.get('details', '')}")
45             fixed=[]
46             for f in v.get("fixed_in", []):
47                 fixed.append(f)
48             lines.append(f"vulnerability fixed in versions: {', '.join(fixed)}")
49             links=[l.get('url', '') for l in v.get('link', []) if isinstance(v.get('link'), list) else []]
50             lines.append("links: " + ", ".join(links))
51             lines.append("-----")
52
53     text="\n".join(lines)
54     fname=f"dataset/{name}.txt"
55     #UnicodeEncodeError: 'charmap' codec can't encode character '\u2192' in position 2112: character maps
56     with open(fname, "w", encoding="utf-8") as f:
57         f.write(text)
58     byte_size=len(text)
59     file_hash=get_sha(fname)
60     source_url = f"https://pypi.org/pypi/{p}/json"
61
62     sources.append({
63         "package": name,
64         "version": p.split("/",1)[1],
65         "url": source_url,
66         "accessed": access_date
67     })
68     manifest.append({
69         "filename": fname.replace("\\", "/"),
70         "byte_size": byte_size,
71         "sha256": file_hash
72     })
73     print(f"{name}.txt, bytes: {len(text)}, number of vuls: {len(vuls)}")
74
75     with open("SOURCES.md", "w", encoding="utf-8") as f:
76         f.write("| Package | Version | Source URL | Accessed | \n")
77         f.write("|-----|-----|-----|-----| \n")
78         for source in sources:
79             f.write(
80                 f"| {source['package']} | {source['version']} | {source['url']} | {source['accessed']} | \n"
81             )
82     print("SOURCES.md created")
83
84     with open("CORPUS_MANIFEST.json", "w", encoding="utf-8") as f:
85         json.dump(manifest, f)
86     print("CORPUS_MANIFEST.json created")
87
88 if __name__=="__main__":
89     main()

```

A. Environment & Setup Install: llama-index, llama-index-embeddings-huggingface, sentence-transformers, faiss-cpu, numpy, pandas.

- Use a public sentence embedding model (e.g., sentence-transformers/all-MiniLM-L6-

v2). (pull from huggingface)

```
class MiniLMEmbedding(BaseEmbedding):
    def __init__(self, model="sentence-transformers/all-MiniLM-L6-v2", **kwargs):
        super().__init__(**kwargs)
        self.model=SentenceTransformer(model)
```

Installations

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932> pip install llama-index
Requirement already satisfied: llama-index in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (0.14.24)
Requirement already satisfied: llama-index-core<0.15.0,>=0.14.24 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index) (0.14.24)
Requirement already satisfied: llama-index-embeddings-openai<0.7,>=0.6.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index) (0.6.0)
Requirement already satisfied: llama-index-llms-openai<0.8,>=0.7.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index) (0.7.10)
Requirement already satisfied: nltk<=3.9.3 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index) (3.10.3)
Requirement already satisfied: aiohttp<4,>=3.8.6 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (3.14.3)
Requirement already satisfied: aiosqlite in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (0.22.1)
Requirement already satisfied: banks<3,>=2.3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (2.5.1)
Requirement already satisfied: dataclasses-json in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (0.6.7)
Requirement already satisfied: deprecated<=1.2.9.3 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (1.3.1)
Requirement already satisfied: dirtyjson<2,>=1.0.8 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (1.0.8)
Requirement already satisfied: filetype<2,>=1.2.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (1.2.0)
Requirement already satisfied: fsspec<=2023.5.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (2026.9.0)
Requirement already satisfied: httpx in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (0.28.1)
Requirement already satisfied: llama-index-workflows<3,>=2.14.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (2.24.1)
Requirement already satisfied: nest-asyncio<2,>=1.5.8 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (1.6.0)
Requirement already satisfied: networkx<=3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (3.6.1)
Requirement already satisfied: numpy in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (2.5.3)
Requirement already satisfied: pillow<=9.0.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (12.3.0)
Requirement already satisfied: platformdirs in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (4.11.11)
Requirement already satisfied: pydantic<=2.8.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (2.13.4)
Requirement already satisfied: pyyaml<=6.0.1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (6.0.3)
Requirement already satisfied: requests<=2.31.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (2.34.2)
Requirement already satisfied: setuptools<=80.9.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (84.0.0)
Requirement already satisfied: sqlalchemy<=1.4.49 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sqlalchemy[asyncio]>=1.4.49->llama-index-core<0.15.0,>=0.14.24->llama-index) (2.0.54)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.2.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (9.1.4)
Requirement already satisfied: tiktoken<=0.7.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (0.14.0)
Requirement already satisfied: tinytag<=2.2.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (2.3.2)
Requirement already satisfied: tqdm<5,>=4.66.1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (4.70.1)
Requirement already satisfied: typing-extensions<=4.5.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (4.16.0)
Requirement already satisfied: typing-inspect<=0.8.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (0.9.0)
Requirement already satisfied: wrapt in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.14.24->llama-index) (2.4.1)
```

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932> pip install llama-index-embeddings-huggingface
Collecting llama-index-embeddings-huggingface
  Downloading llama_index_embeddings_huggingface-0.8.0-py3-none-any.whl.metadata (459 bytes)
Requirement already satisfied: huggingface-hub<0.19.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-embeddings-huggingface) (0.19.0)
Requirement already satisfied: llama-index-core<0.15.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-embeddings-huggingface) (0.14.24)
Requirement already satisfied: sentence-transformers>=2.6.1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-embeddings-huggingface) (6.1.0)
Requirement already satisfied: click<9.0.0,>=8.4.2 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<0.19.0>huggingface-hub[inference]>0.19.0->llama-index-embeddings-huggingface) (8.5.0)
Requirement already satisfied: filelock>=3.10.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<0.19.0>huggingface-hub[inference]>0.19.0->llama-index-embeddings-huggingface) (4.0.1)
Requirement already satisfied: fsspec>=2023.5.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<0.19.0>huggingface-hub[inference]>0.19.0->llama-index-embeddings-huggingface) (2026.9.0)
Requirement already satisfied: hf-xet<0.0.0,>=1.5.2 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<0.19.0>huggingface-hub[inference]>0.19.0->llama-index-embeddings-huggingface) (1.6.0)
Requirement already satisfied: httpx<1,>=0.23.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<0.19.0>huggingface-hub[inference]>0.19.0->llama-index-embeddings-huggingface) (0.28.1)
Requirement already satisfied: packaging>=20.9 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<0.19.0>huggingface-hub[inference]>0.19.0->llama-index-embeddings-huggingface) (26.3)
Requirement already satisfied: pyyaml>=5.1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<0.19.0>huggingface-hub[inference]>0.19.0->llama-index-embeddings-huggingface) (6.0.3)
Requirement already satisfied: tqdm>=4.42.1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<0.19.0>huggingface-hub[inference]>0.19.0->llama-index-embeddings-huggingface) (4.70.1)
Requirement already satisfied: typing-extensions>=4.1.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<0.19.0>huggingface-hub[inference]>0.19.0->llama-index-embeddings-huggingface) (4.16.0)
WARNING: huggingface-hub 1.32.0 does not provide the extra 'inference'
Requirement already satisfied: aiohttp<4,>=3.8.6 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (3.14.3)
Requirement already satisfied: aiosqlite in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (0.22.1)
Requirement already satisfied: banks<3,>=2.3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (2.5.1)
Requirement already satisfied: dataclasses-json in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (0.6.7)
Requirement already satisfied: deprecated>=1.2.9.3 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (1.3.1)
Requirement already satisfied: dirtyjson<2,>=1.0.8 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (1.0.8)
Requirement already satisfied: filetype<2,>=1.2.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (1.2.0)
Requirement already satisfied: llama-index-workflows<3,>=2.14.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (2.24.1)
Requirement already satisfied: nest-asyncio<2,>=1.5.8 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (1.6.0)
Requirement already satisfied: networkx<=3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (3.6.1)
Requirement already satisfied: nltk<=3.9.3 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (3.10.3)
Requirement already satisfied: numpy in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (2.5.3)
Requirement already satisfied: pillow>=9.0.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (12.3.0)
Requirement already satisfied: platformdirs in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (4.11.11)
Requirement already satisfied: pydantic>=2.8.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (2.13.4)
Requirement already satisfied: requests>=2.31.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (2.34.2)
Requirement already satisfied: setuptools>=80.9.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (84.0.0)
Requirement already satisfied: sqlalchemy>=1.4.49 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sqlalchemy[asyncio]>=1.4.49->llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (2.0.54)
Requirement already satisfied: tenacity<=8.4.0,<10.0.0,>=8.2.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (9.1.4)
Requirement already satisfied: tiktoken>=0.7.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (0.14.0)
Requirement already satisfied: tinytag>=2.2.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (2.3.2)
Requirement already satisfied: typing-inspect>=0.8.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (0.9.0)
Requirement already satisfied: wrapt in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-core<0.15.0,>=0.13.0->llama-index-embeddings-huggingface) (1.17.0)
```

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932> pip install llama-index-embeddings-huggingface
Requirement already satisfied: griffe in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from banks<3,>=2.3.0->llama-index-embeddings-huggingface) (2.3.0)
Requirement already satisfied: Jinja2 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from banks<3,>=2.3.0->llama-index-embeddings-huggingface) (3.1.6)
Requirement already satisfied: anyio in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1,>=0.23.0->huggingface-hub[Inference]>=0.19.0->llama-index-embeddings-huggingface) (4.14.2)
Requirement already satisfied: certifi in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1,>=0.23.0->huggingface-hub[Inference]>=0.19.0->llama-index-embeddings-huggingface) (2026.7.22)
Requirement already satisfied: httpcore==1.* in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1,>=0.23.0->huggingface-hub[Inference]>=0.19.0->llama-index-embeddings-huggingface) (1.0.9)
Requirement already satisfied: idna in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1,>=0.23.0->huggingface-hub[Inference]>=0.19.0->llama-index-embeddings-huggingface) (3.19)
Requirement already satisfied: h11<0.16 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from httpcore==1.*->httpx>huggingface-hub[Inference]>=0.19.0->llama-index-embeddings-huggingface) (0.16.0)
Requirement already satisfied: llama-index-instrumentation<0.4.3 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from llama-index-embeddings-huggingface) (0.7.1)
Requirement already satisfied: defusedxml in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from nltk<3.9.3->llama-index-embeddings-huggingface) (0.7.1)
Requirement already satisfied: joblib in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from nltk<3.9.3->llama-index-embeddings-huggingface) (1.6.0)
Requirement already satisfied: regex<2021.8.3 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from nltk<3.9.3->llama-index-embeddings-huggingface) (2026.9.10)
Requirement already satisfied: annotated-types<0.6.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from pydantic<2.10.0->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.4 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from pydantic<2.10.0->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (2.46.4)
Requirement already satisfied: typing-inspection<0.4.2 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from pydantic<2.10.0->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (0.4.4)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from requests<2.32.0->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (3.5.1)
Requirement already satisfied: urllib3<3,>=1.26 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from requests<2.32.0->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (2.7.0)
Requirement already satisfied: narwhals<2.0.1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from scikit-learn<1.6.0->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (2.26.0)
Requirement already satisfied: threadpoolctl<3.5.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from scikit-learn<1.6.0->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (3.7.0)
Requirement already satisfied: greenlet<1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sqlalchemy<1.4.49->[asyncio]>=1.4.49->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (3.5.6)
Requirement already satisfied: sympy<1.13.3 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from torch>=2.2->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (1.14.0)
Requirement already satisfied: colorama in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from tqdm<=4.42.1->huggingface-hub[Inference]>=0.19.0->llama-index-embeddings-huggingface) (0.4.6)
Requirement already satisfied: typer in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from transformers<6.0.0,>=5.0.0->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (0.27.2)
Requirement already satisfied: safetensors<0.8.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from transformers<6.0.0->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (0.8.0)
Requirement already satisfied: mpy-extensions<0.3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from typing-extensions<4.13.0->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (1.1.0)
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from dataclasses-json<0.6.7->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (3.26.2)
Requirement already satisfied: cloudpickle<3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from joblib->nltk<3.9.3->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (3.1.2)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sympy>=1.13.0->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (1.3.0)
Requirement already satisfied: griffecli==2.3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from griffe->banks<3.1.0->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (2.3.0)
Requirement already satisfied: griffe==2.3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from griffe->banks<3.1.0->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (2.3.0)
Requirement already satisfied: MarkupSafe==2.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from Jinja2->banks<3.1.0->llama-index-core<0.15,>=0.13.0->llama-index-embeddings-huggingface) (3.0.3)
Requirement already satisfied: shellingham==1.3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from typer->transformers<6.0.0,>=5.0.0->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (1.5.4)
Requirement already satisfied: rich==13.8.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from typer->transformers<6.0.0,>=5.0.0->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (15.0.0)
Requirement already satisfied: annotated-doc<0.0.2 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from typer->transformers<6.0.0,>=5.0.0->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (0.0.5)
Requirement already satisfied: markdown-it-py==2.2.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from rich>=13.8.0->transformers<6.0.0,>=5.0.0->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from rich>=13.8.0->transformers<6.0.0,>=5.0.0->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (2.21.0)
Requirement already satisfied: mdurl==0.1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from markdown-it-py==2.2.0->transformers<6.0.0,>=5.0.0->sentence-transformers<2.6.1->llama-index-embeddings-huggingface) (0.1.2)
Downloading llama_index_embeddings_huggingface-0.8.0-py3-none-any.whl (8.9 kB)
Installing collected packages: llama-index-embeddings-huggingface
Successfully installed llama-index-embeddings-huggingface-0.8.0
```

```

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932> pip install sentence_transformers
Requirement already satisfied: sentence_transformers in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (6.1.0)
Requirement already satisfied: transformers<6.0.0,>=5.0.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sentence_transformers) (5.17.0)
Requirement already satisfied: tokenizers==0.19 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sentence_transformers) (0.23.2)
Requirement already satisfied: huggingface-hub<2.0.0,>=1.3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sentence_transformers) (1.32.0)
Requirement already satisfied: torch==2.2 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sentence_transformers) (2.14.0)
Requirement already satisfied: numpy>=1.24.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sentence_transformers) (2.5.3)
Requirement already satisfied: scikit-learn==1.1.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sentence_transformers) (1.9.1)
Requirement already satisfied: scipy>=1.0.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sentence_transformers) (1.18.1)
Requirement already satisfied: typing_extensions>=4.10.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sentence_transformers) (4.16.0)
Requirement already satisfied: tqdm==4.0.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sentence_transformers) (4.70.1)
Requirement already satisfied: click<9.0.0,>=8.4.2 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (8.5.0)
Requirement already satisfied: filelock>=3.10.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (4.0.1)
Requirement already satisfied: fsspec>=2023.5.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (2026.9.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.2 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (1.6.0)
Requirement already satisfied: httpx<1,>=0.23.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (0.28.1)
Requirement already satisfied: packaging>=20.9 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (26.3)
Requirement already satisfied: pyyaml>=5.1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (6.0.3)
Requirement already satisfied: joblib>=1.4.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from scikit-learn==1.1.0->sentence_transformers) (1.6.0)
Requirement already satisfied: narwhals>=2.0.1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from scikit-learn==1.1.0->sentence_transformers) (2.26.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from scikit-learn==1.1.0->sentence_transformers) (3.7.0)
Requirement already satisfied: setuptools>=77.0.3 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from torch==2.2->sentence_transformers) (84.0.0)
Requirement already satisfied: sympy>=1.13.3 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from torch==2.2->sentence_transformers) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from torch==2.2->sentence_transformers) (3.6.1)
Requirement already satisfied: Jinja2 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from torch==2.2->sentence_transformers) (3.1.6)
Requirement already satisfied: colorama in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from tqdm==4.0.0->sentence_transformers) (0.4.6)
Requirement already satisfied: regex>=2025.10.22 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from transformers<6.0.0,>=5.0.0->sentence_transformers) (2026.9.10)
Requirement already satisfied: typer in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from transformers<6.0.0,>=5.0.0->sentence_transformers) (0.27.2)
Requirement already satisfied: safetensors>=0.8.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from transformers<6.0.0,>=5.0.0->sentence_transformers) (0.8.0)
Requirement already satisfied: anyio in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (4.14.2)
Requirement already satisfied: certifi in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (2026.7.22)
Requirement already satisfied: httpcore==1.* in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (1.0.9)
Requirement already satisfied: idna in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (3.19)
Requirement already satisfied: h11>=0.16 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0->sentence_transformers) (0.16.0)
Requirement already satisfied: cloudpickle>=3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from joblib>=1.4.0->scikit-learn==1.1.0->sentence_transformers) (3.1.2)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from sympy>=1.13.3->torch==2.2->sentence_transformers) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from Jinja2->torch==2.2->sentence_transformers) (3.0.3)
Requirement already satisfied: shellingham>=1.3.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from typer->transformers<6.0.0,>=5.0.0->sentence_transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from typer->transformers<6.0.0,>=5.0.0->sentence_transformers) (15.0.0)
Requirement already satisfied: annotated-doc>=0.0.2 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from typer->transformers<6.0.0,>=5.0.0->sentence_transformers) (0.1.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from rich>=13.8.0->transformers<6.0.0,>=5.0.0->sentence_transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from rich>=13.8.0->transformers<6.0.0,>=5.0.0->sentence_transformers) (2.21.0)
Requirement already satisfied: mdurl>=0.1 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from markdown-it-py>=2.2.0->typer->transformers<6.0.0,>=5.0.0->sentence_transformers) (0.1.2)

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip

```

```

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932> pip install faiss-cpu
Collecting faiss-cpu
  Downloading faiss_cpu-1.15.1-cp312-cp312-win_amd64.whl.metadata (7.8 kB)
Requirement already satisfied: numpy>=1.25 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from faiss-cpu) (2.5.3)
Requirement already satisfied: packaging in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from faiss-cpu) (26.3)
Downloading faiss_cpu-1.15.1-cp312-cp312-win_amd64.whl (16.3 MB)
 16.3/16.3 MB 68.5 MB/s eta 0:00:00
Installing collected packages: faiss-cpu
Successfully installed faiss-cpu-1.15.1

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932> pip install numpy
Requirement already satisfied: numpy in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (2.5.3)

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932> pip install pandas
Collecting pandas
  Downloading pandas-3.0.6-cp312-cp312-win_amd64.whl.metadata (19 kB)
Requirement already satisfied: numpy>=1.26.0 in c:\users\natal\appdata\local\programs\python\python312\lib\site-packages (from pandas) (2.5.3)
Collecting python-dateutil>=2.8.2 (from pandas)
  Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting tzdata (from pandas)
  Downloading tzdata-2026.4-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
  Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading pandas-3.0.6-cp312-cp312-win_amd64.whl (9.7 MB)
 9.7/9.7 MB 35.3 MB/s eta 0:00:00
Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading tzdata-2026.4-py2.py3-none-any.whl (347 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Installing collected packages: tzdata, six, python-dateutil, pandas
Successfully installed pandas-3.0.6 python-dateutil-2.9.0.post0 six-1.17.0 tzdata-2026.4

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip

```

B. One retrieval-only pipeline per technique

For each chunker (Token / Semantic / Sentence-window):

1. Chunking

- Token: set a token chunk_size and chunk_overlap (choose sensible values). (LlamaIndex)

```

28 #different chunking
29 def token_node(docs):
30     splitter=TokenTextSplitter(chunk_size=200,chunk_overlap=40)
31     return splitter.get_nodes_from_documents(docs)
32

```

- Semantic: pick a buffer_size and use your embed model for the splitter to find semantically coherent boundaries. (LlamaIndex)

```

33 def semantic_node(docs):
34     splitter=SemanticSplitterNodeParser(buffer_size=1, breakpoint_percentile_threshold=90, embed_model=model)
35     return splitter.get_nodes_from_documents(docs)
36

```

- Sentence-window: split to single sentences and attach a window (neighbor sentences) in metadata to keep surrounding context available. (LlamaIndex)

```

37 def sentence_window_node(docs):
38     splitter=SentenceWindowNodeParser.from_defaults(window_size=3, window_metadata_key="window", original_text_metadata_key="original_text")
39     return splitter.get_nodes_from_documents(docs)
40
41 techniques= {"token": token_node, "semantic": semantic_node, "sentenceWindow": sentence_window_node}

```

2. Indexing (in memory)

- Build a VectorStoreIndex over your nodes with an in-memory vector store (e.g., SimpleVectorStore) to keep everything local and fast. (LlamaIndex)

```

102     for name, build in techniques.items():
103         print(f"\n{name} index")
104         nodes=build(doc)
105         lengths=[len(n.get_content()) for n in nodes]
106         chunk[name]= {"num_chunks": len(nodes), "avg_chunk_len":sum(lengths)/len(lengths)}
107         print(f"{name}:{len(nodes)} chunks, avg length {chunk[name]['avg_chunk_len']:.0f} chars")
108
109     index = VectorStoreIndex(nodes)

```

3. Retrieval-only function

Write a helper that, given a query and k, does the following:

- Compute the query embedding (show its dimension and the first 8 values).

```

51     #for top k nodes, auto to top 3
52     def run_query(technique, index, query, k=3):
53         q=model.get_query_embedding(query)
54         start = time.perf_counter()
55         retriever = index.as_retriever(similarity_top_k=k)
56         result = retriever.retrieve(query)
57         latency = (time.perf_counter()-start)*1000
58         rows=[]
59         chunk_embeddings=[]
60         for rank, node in enumerate(result, start=1):
61             text=node.get_content()
62             chunk_embedding=model.get_text_embedding(text)
63             chunk_embeddings.append(chunk_embedding)
64             rows.append({
65                 "technique": technique,
66                 "query": query,
67                 "rank": rank,
68                 "store_score": float(node.score) if node.score is not None else None,
69                 "cosine_sim": cosine(q,chunk_embedding),
70                 "chunk_len":len(text),
71                 #first ~160 characters of text preview
72                 "preview": text[:160].replace("\n", " "),
73                 "source_file": node.node.metadata.get("file_name", "")
74             })
75         matrix=np.array(chunk_embeddings) if chunk_embeddings else np.zeros((0,len(q)))
76         print(f"\n{technique} | query: {query!r}")
77         print(f"query vector shape: {np.array(q).shape}, first 8 values: {np.round(q[:8], 3)}")

```

- Retrieve top-k nodes; for each, compute and print:

- Store similarity score (if available from retriever).

```
55 retriever = index.as_retriever(similarity_top_k=k)
56 result = retriever.retrieve(query)
57 latency = (time.perf_counter()-start)*1000
58 rows=[]
59 chunk_embeddings=[]
60 for rank, node in enumerate(result, start=1):
61     text=node.node.get_content()
62     chunk_embedding=model.get_text_embedding(text)
63     chunk_embeddings.append(chunk_embedding)
64     rows.append({
65         "technique": technique,
66         "query": query,
67         "rank": rank,
68         "store_score": float(node.score) if node.score is not None else None,
69         "cosine_sim": cosine(q,chunk_embedding),
70         "chunk_len":len(text),
71         #first ~160 characters of text preview
72         "preview": text[:160].replace("\n", " "),
73         "source_file": node.node.metadata.get("file_name", "")
74     })
```

- Cosine similarity between the query embedding and the document embedding (compute embeddings of the returned chunks explicitly).

```

51 #for top k nodes, auto to top 3
52 def run_query(technique, index, query, k=3):
53     q=model.get_query_embedding(query)
54     start = time.perf_counter()
55     retriever = index.as_retriever(similarity_top_k=k)
56     result = retriever.retrieve(query)
57     latency = (time.perf_counter()-start)*1000
58     rows=[]
59     chunk_embeddings=[]
60     for rank, node in enumerate(result, start=1):
61         text=node.node.get_content()
62         chunk_embedding=model.get_text_embedding(text)
63         chunk_embeddings.append(chunk_embedding)
64         rows.append({
65             "technique": technique,
66             "query": query,
67             "rank": rank,
68             "store_score": float(node.score) if node.score is not None else None,
69             "cosine_sim": cosine(q,chunk_embedding),
70             "chunk_len":len(text),
71             #first ~160 characters of text preview
72             "preview": text[:160].replace("\n", " "),
73             "source_file": node.node.metadata.get("file_name", "")
74         })
75     matrix=np.array(chunk_embeddings) if chunk_embeddings else np.zeros((0,len(q)))
76     print(f"\n{technique} | query: {query!r}")
77     print(f"query vector shape: {np.array(q).shape}, first 8 values: {np.round(q[:8], 3)}")
78     print(f"stacked vectors shape: {matrix.shape}")
79     #print table for each technique
80     print("rank, store_score, cosine_sim, chunk_len, preview")
81     for r in rows:
82         s=f"{r['store_score']:.3f}" if r["store_score"] is not None else "none"
83         print(f"{r['rank']}, {s}, {r['cosine_sim']}, {r['chunk_len']}, {r['preview']}")
84
85     return rows, latency

```

- Chunk length and a short text preview (first ~160 chars).
- Print the shapes of the query vector and the stacked doc vectors.

```

70         "chunk_len":len(text),
71         #first ~160 characters of text preview
72         "preview": text[:160].replace("\n", " "),
73         "source_file": node.node.metadata.get("file_name", "")
74     })
75     matrix=np.array(chunk_embeddings) if chunk_embeddings else np.zeros((0,len(q)))
76     print(f"\n{technique} | query: {query!r}")
77     print(f"query vector shape: {np.array(q).shape}, first 8 values: {np.round(q[:8], 3)}")
78     print(f"stacked vectors shape: {matrix.shape}")
79     #print table for each technique
80     print("rank, store_score, cosine_sim, chunk_len, preview")
81     for r in rows:
82         s=f"{r['store_score']:.3f}" if r["store_score"] is not None else "none"
83         print(f"{r['rank']}, {s}, {r['cosine_sim']}, {r['chunk_len']}, {r['preview']}")
84
85     return rows, latency

```

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports>python .\retrieval_pipeline.py
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
Loading weights: 100%|██████████████████████████████████████████████████████████████████████████████| 103/103 [00:00<00:00, 9249.63it/s]

token index
token:438 chunks, avg length 663 chars

token | query: 'What PyYAML method could allow arbitrary code execution and when was it fixed?'
query vector shape: (384,), first 8 values: [-0.043  0.031 -0.079 -0.009 -0.045 -0.063 -0.113 -0.036]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.738, 0.7237804265392968, 804, could use this flaw to execute arbitrary code on the system by abusing the python/object/new
constructor. vulnerability fixed in versions: 5.3.1 links: -----
2, 0.727, 0.7264973669635378, 702, package: PyYAML (version:pyyaml/5.3) vulnerability count: 4 id: PYSEC-2020-96 aliases: CVE-
2020-1747, CVE-2025-50460, GHSA-6757-jp84-gxfx, GHSA-fm6c-f59h-7mmg
3, 0.723, 0.6947960142607781, 796, attacker to execute arbitrary code on the system by abusing the python/object/new constructo
r. This flaw is due to an incomplete fix for CVE-2020-1747. vulnerab

token | query: "What bug affected Paramiko's write_private_key_file?"
query vector shape: (384,), first 8 values: [-0.04 -0.006 -0.085  0.032  0.032 -0.03 -0.02  0.095]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.750, 0.7290187621460011, 530, package: paramiko (version:paramiko/2.4.1) vulnerability count: 4 id: PYSEC-2022-166 aliase
s: CVE-2022-24302, GHSA-f8q4-jwmm-wx3wv summary: None details: In Par
2, 0.567, 0.5682809410690168, 491, None details: Paramiko version 2.4.1, 2.3.2, 2.2.3, 2.1.5, 2.0.8, 1.18.5, 1.17.6 contains a
Incorrect Access Control vulnerability in SSH server that can result
3, 0.545, 0.4699150282476977, 426, PYSEC-2026-2858 summary: None details: In Paramiko through 4.0.0 before a448945, rsakey.py a
llows the SHA-1 algorithm. vulnerability fixed in versions: links:

token | query: 'What NumPy component had a string comparision bug and when was it fixed?'
query vector shape: (384,), first 8 values: [-0.042 -0.016 -0.035  0.003 -0.001 -0.046 -0.033  0.038]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.644, 0.6877965828168188, 376, package: numpy (version:numpy/1.21.0) vulnerability count: 1 id: GHSA-fpfv-jqm9-f5jm alias
es: CVE-2021-34141, PYSEC-2021-855 summary: None details: Incomplete
2, 0.384, 0.4082623583283577, 607, fixed in versions: 12.2.0 links: ----- id: PYSEC-2026-165 aliases: CV
E-2026-42308, GHSA-wjx4-4jcj-g98j summary: None details: Pillow is a
3, 0.374, 0.3880033653290349, 875, small subgroup. Only SECT curves are impacted by this. ## Credit This vulnerability was d
iscovered by: - XlabAI Team of Tencent Xuanwu Lab - Atuin Automated

token | query: 'Which package has the most vulnerabilites in the datasets?'
query vector shape: (384,), first 8 values: [-0.051  0.006 -0.069 -0.05  0.05 -0.032 -0.009  0.08 ]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.386, 0.42865609164071083, 376, package: numpy (version:numpy/1.21.0) vulnerability count: 1 id: GHSA-fpfv-jqm9-f5jm alias
es: CVE-2021-34141, PYSEC-2021-855 summary: None details: Incomplete
2, 0.366, 0.3765998807712292, 597, package: pillow (version:pillow/8.2.0) vulnerability count: 52 id: PYSEC-2021-331 aliases:
CVE-2021-34552, GHSA-7534-mm45-c74v summary: None details: Pillow th
3, 0.356, 0.39722910619797513, 574, BIT-pillow-2022-45198, GVE-2022-45198, GHSA-m2vv-5vj5-2hm7 summary: None details: Pillow be
fore 9.2.0 performs Improper Handling of Highly Compressed GIF Data (

token | query: 'Name two packages with RCE related vulnerabilites in SSH or networking code.'
query vector shape: (384,), first 8 values: [-0.078  0.052 -0.054 -0.051  0.081  0.005 -0.009  0.099]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.531, 0.5330640997602768, 491, None details: Paramiko version 2.4.1, 2.3.2, 2.2.3, 2.1.5, 2.0.8, 1.18.5, 1.17.6 contains a
Incorrect Access Control vulnerability in SSH server that can result
2, 0.384, 0.40500948641356066, 519, site) reactor.run() ```` Output: ```` > curl1 -H"Host:<h1>HELLO THERE</h1>" http://localhost:8
080/ <html> <head><title>404 - No Such Resource</title></head> <
3, 0.351, 0.37791884441974266, 376, package: numpy (version:numpy/1.21.0) vulnerability count: 1 id: GHSA-fpfv-jqm9-f5jm alias
es: CVE-2021-34141, PYSEC-2021-855 summary: None details: Incomplete

semantic index
semantic:118 chunks, avg length 1972 chars

semantic | query: 'What PyYAML method could allow arbitrary code execution and when was it fixed?'
query vector shape: (384,), first 8 values: [-0.043  0.031 -0.079 -0.009 -0.045 -0.063 -0.113 -0.036]
stacked vectors shape: (3, 384)
rank, store score, cosine sim, chunk len, preview
```

```

1, 0.731, 0.6867692839036744, 845, This flaw allows an attacker to execute arbitrary code on the system by abusing the python/o
bject/new constructor. This flaw is due to an incomplete fix for CVE
2, 0.724, 0.726753249451751, 1107, package: PyYAML (version:pyyaml/5.3) vulnerability count: 4 id: PYSEC-2020-96 aliases: CVE-
2020-1747, CVE-2025-50460, GHSA-6757-jp84-gxfx, GHSA-fm6c-f59h-7mmg
3, 0.710, 0.6335310828338969, 653, This flaw allows an attacker to execute arbitrary code on the system by abusing the python/o
bject/new constructor. This flaw is due to an incomplete fix for CVE

semantic | query: "What bug affected Paramiko's write_private_key_file?"
query vector shape: (384,), first 8 values: [-0.04 -0.006 -0.085 0.032 0.032 -0.03 -0.02 0.095]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.733, 0.7150091588376839, 704, package: paramiko (version:paramiko/2.4.1) vulnerability count: 4 id: PYSEC-2022-166 aliase
s: CVE-2022-24302, GHSA-f8q4-jwww-x3wv summary: None details: In Par
2, 0.575, 0.5331738596991491, 578, vulnerability fixed in versions: 2.4.2, 2.3.3, 2.2.4, 2.1.6, 2.0.9 links: -----
----- id: GHSA-r374-rxx8-8654 aliases: CVE-2026-44405, PYSEC-20
3, 0.430, 0.4013753060649975, 2676, This can lead to security issues in various situations, such as the most commonly used sign
ature verification (ECDSA) and shared key negotiation (ECDH). When th

semantic | query: 'What NumPy component had a string comparision bug and when was it fixed?'
query vector shape: (384,), first 8 values: [-0.042 -0.016 -0.035 0.003 -0.001 -0.046 -0.033 0.038]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.644, 0.6877965828168188, 376, package: numpy (version:numpy/1.21.0) vulnerability count: 1 id: GHSA-fpfv-jqm9-f5jm alias
s: CVE-2021-34141, PYSEC-2021-855 summary: None details: Incomplete
2, 0.358, 0.3496219164474602, 653, This flaw allows an attacker to execute arbitrary code on the system by abusing the python/o
bject/new constructor. This flaw is due to an incomplete fix for CVE
3, 0.356, 0.3502011060755762, 6436, ### Patches Patched version: 12.2.0. PdfParser (introduced in Pillow 4.2.0) follows Prev p
ointers in PDF trailers to read cross-reference sections. If a traile

semantic | query: 'Which package has the most vulnerabilites in the datasets?'
query vector shape: (384,), first 8 values: [-0.051 0.006 -0.069 -0.05 0.05 -0.032 -0.009 0.08 ]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.386, 0.42865609164071083, 376, package: numpy (version:numpy/1.21.0) vulnerability count: 1 id: GHSA-fpfv-jqm9-f5jm alias
es: CVE-2021-34141, PYSEC-2021-855 summary: None details: Incomplete
2, 0.347, 0.35019274272291423, 6220, package: pillow (version:pillow/8.2.0) vulnerability count: 52 id: PYSEC-2021-331 aliases
: CVE-2021-34552, GHSA-7534-mm45-c74v summary: None details: Pillow th
3, 0.337, 0.3653976498432271, 1866, package: Django (version:django/1.11.28) vulnerability count: 18 id: PYSEC-2020-36 aliases
: BIT-django-2020-9402, CVE-2020-9402, GHSA-3gh2-xw74-jmcw, PYSEC-202

semantic | query: 'Name two packages with RCE related vulnerabilites in SSH or networking code.'
query vector shape: (384,), first 8 values: [-0.078 0.052 -0.054 -0.051 0.081 0.005 -0.009 0.099]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.387, 0.3505041270339254, 3193, More details about the vulnerability itself can be found in https://www.openssl.org/news/se
cadv/20230530.txt. If you are building cryptography source ("sdist")
2, 0.380, 0.39146292665432536, 2856, More details about the vulnerabilities themselves can be found in https://www.openssl.org/
news/secadv/20221213.txt and https://www.openssl.org/news/secadv/20230
3, 0.354, 0.3454630812245813, 578, vulnerability fixed in versions: 2.4.2, 2.3.3, 2.2.4, 2.1.6, 2.0.9 links: -----
----- id: GHSA-r374-rxx8-8654 aliases: CVE-2026-44405, PYSEC-20

sentenceWindow index
sentenceWindow:1031 chunks, avg length 226 chars

sentenceWindow | query: 'What PyYAML method could allow arbitrary code execution and when was it fixed?'
query vector shape: (384,), first 8 values: [-0.043 0.031 -0.079 -0.009 -0.045 -0.063 -0.113 -0.036]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.712, 0.7256470891631759, 413, package: PyYAML (version:pyyaml/5.3) vulnerability count: 4 id: PYSEC-2020-96 aliases: CVE-
2020-1747, CVE-2025-50460, GHSA-6757-jp84-gxfx, GHSA-fm6c-f59h-7mmg
2, 0.707, 0.4460866878703384, 115, This flaw allows an attacker to execute arbitrary code on the system by abusing the python/o
bject/new constructor.
3, 0.707, 0.4460866878703384, 115, This flaw allows an attacker to execute arbitrary code on the system by abusing the python/o
bject/new constructor.

sentenceWindow | query: "What bug affected Paramiko's write_private_key_file?"
query vector shape: (384,), first 8 values: [-0.04 -0.006 -0.085 0.032 0.032 -0.03 -0.02 0.095]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.789, 0.7711990802247894, 315, package: paramiko (version:paramiko/2.4.1) vulnerability count: 4 id: PYSEC-2022-166 aliase

```

```

s: CVE-2022-24302, GHSA-f8q4-jvww-x3wv summary: None details: In Par
2, 0.641, 0.32539294028084276, 69, vulnerability fixed in versions: links: -----
3, 0.622, 0.5613412682848823, 271, vulnerability fixed in versions: 2.4.2, 2.3.3, 2.2.4, 2.1.6, 2.0.9 links: -----
----- id: GHSA-r374-rxx8-8654 aliases: CVE-2026-44405, PYSEC-20

sentenceWindow | query: 'What NumPy component had a string comparision bug and when was it fixed?'
query vector shape: (384,), first 8 values: [-0.042 -0.016 -0.035  0.003 -0.001 -0.046 -0.033  0.038]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.626, 0.6595318335982803, 303, package: numpy (version:numpy/1.21.0) vulnerability count: 1 id: GHSA-fpfv-jqm9-f5jm alias
s: CVE-2021-34141, PYSEC-2021-855 summary: None details: Incomplete
2, 0.582, 0.262929101599153, 73, vulnerability fixed in versions: 1.22 links: -----
3, 0.373, 0.36884570835772607, 193, The bug is reachable through the public save API: ``python im.save(out, format="TGA", com
pression="tga_rle") `` Older affected Pillow versions use the equiv

sentenceWindow | query: 'Which package has the most vulnerabilites in the datasets?'
query vector shape: (384,), first 8 values: [-0.051  0.006 -0.069 -0.05  0.05 -0.032 -0.009  0.08 ]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.427, 0.4847990656467272, 303, package: numpy (version:numpy/1.21.0) vulnerability count: 1 id: GHSA-fpfv-jqm9-f5jm alias
s: CVE-2021-34141, PYSEC-2021-855 summary: None details: Incomplete
2, 0.402, 0.4495544475793688, 372, package: Django (version:django/1.11.28) vulnerability count: 18 id: PYSEC-2020-36 aliases:
BIT-django-2020-9402, CVE-2020-9402, GHSA-3gh2-xw74-jmcw, PYSEC-202
3, 0.352, 0.34902080227094245, 268, vulnerability fixed in versions: 42.0.2 links: ----- id: PYSEC-2026-
2141 aliases: CVE-2026-26007, GHSA-r6ph-v2qm-q3c2 summary: None detai

sentenceWindow | query: 'Name two packages with RCE related vulnerabilites in SSH or networking code.'
query vector shape: (384,), first 8 values: [-0.078  0.052 -0.054 -0.051  0.081  0.005 -0.009  0.099]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.461, 0.4515095687466853, 326, vulnerability fixed in versions: 2.9.3, 2.10.1 links: ----- id: GHSA-
f2j6-wrhh-v25m aliases: CVE-2018-1000805, PYSEC-2018-69 summary: Non
2, 0.386, 0.42900134835625914, 117, More details about the vulnerabilities themselves can be found in https://www.openssl.org/n
ews/secadv/20230908.txt.
3, 0.383, 0.3991560332907133, 170, More details about the vulnerabilities themselves can be found in https://www.openssl.org/ne
ws/secadv/20221213.txt and https://www.openssl.org/news/secadv/20230

```

Your printed output should clearly identify the technique used and list a table with: rank, store_score, cosine_sim, chunk_len, preview. Save this per-query, per-technique output as machine-readable data (CSV/JSON/JSONL) in raw/, along with a script that recomputes the summary tables below from it.

The following are the raw files printed from running the above output console screenshot

```

data260-0932 > reports > hw03 > raw > {} per_query_results.json > ...
1 {"technique": "token", "query": "What PyYAML method could allow arbitrary code execution and when was it fixed?", "rank": 1, "store_score": 0.7504473933474267, "cosine": 0.9999999999999999}
2 {"technique": "token", "query": "What PyYAML method could allow arbitrary code execution and when was it fixed?", "rank": 2, "store_score": 0.7504473933474267, "cosine": 0.9999999999999999}
3 {"technique": "token", "query": "What PyYAML method could allow arbitrary code execution and when was it fixed?", "rank": 3, "store_score": 0.7504473933474267, "cosine": 0.9999999999999999}
4 {"technique": "token", "query": "What bug affected Paramiko's write_private_key_file?", "rank": 1, "store_score": 0.7504473933474267, "cosine": 0.9999999999999999}
5 {"technique": "token", "query": "What bug affected Paramiko's write_private_key_file?", "rank": 2, "store_score": 0.5666308442881579, "cosine": 0.9999999999999999}
6 {"technique": "token", "query": "What bug affected Paramiko's write_private_key_file?", "rank": 3, "store_score": 0.5454592137383141, "cosine": 0.9999999999999999}
7 {"technique": "token", "query": "What NumPy component had a string comparison bug and when was it fixed?", "rank": 1, "store_score": 0.643071, "cosine": 0.9999999999999999}
8 {"technique": "token", "query": "What NumPy component had a string comparison bug and when was it fixed?", "rank": 2, "store_score": 0.38356, "cosine": 0.9999999999999999}
9 {"technique": "token", "query": "What NumPy component had a string comparison bug and when was it fixed?", "rank": 3, "store_score": 0.37373, "cosine": 0.9999999999999999}
10 {"technique": "token", "query": "Which package has the most vulnerabilities in the datasets?", "rank": 1, "store_score": 0.3858228579573216, "cosine": 0.9999999999999999}
11 {"technique": "token", "query": "Which package has the most vulnerabilities in the datasets?", "rank": 2, "store_score": 0.3659654600977158, "cosine": 0.9999999999999999}
12 {"technique": "token", "query": "Which package has the most vulnerabilities in the datasets?", "rank": 3, "store_score": 0.3561612971373188, "cosine": 0.9999999999999999}
13 {"technique": "token", "query": "Name two packages with RCE related vulnerabilities in SSH or networking code.", "rank": 1, "store_score": 0.64, "cosine": 0.9999999999999999}
14 {"technique": "token", "query": "Name two packages with RCE related vulnerabilities in SSH or networking code.", "rank": 2, "store_score": 0.64, "cosine": 0.9999999999999999}
15 {"technique": "token", "query": "Name two packages with RCE related vulnerabilities in SSH or networking code.", "rank": 3, "store_score": 0.64, "cosine": 0.9999999999999999}
16 {"technique": "semantic", "query": "What PyYAML method could allow arbitrary code execution and when was it fixed?", "rank": 1, "store_score": 0.7504473933474267, "cosine": 0.9999999999999999}
17 {"technique": "semantic", "query": "What PyYAML method could allow arbitrary code execution and when was it fixed?", "rank": 2, "store_score": 0.7504473933474267, "cosine": 0.9999999999999999}
18 {"technique": "semantic", "query": "What PyYAML method could allow arbitrary code execution and when was it fixed?", "rank": 3, "store_score": 0.7504473933474267, "cosine": 0.9999999999999999}
19 {"technique": "semantic", "query": "What bug affected Paramiko's write_private_key_file?", "rank": 1, "store_score": 0.7331680209282149, "cosine": 0.9999999999999999}
20 {"technique": "semantic", "query": "What bug affected Paramiko's write_private_key_file?", "rank": 2, "store_score": 0.5754923212128946, "cosine": 0.9999999999999999}
21 {"technique": "semantic", "query": "What bug affected Paramiko's write_private_key_file?", "rank": 3, "store_score": 0.4302944037611603, "cosine": 0.9999999999999999}
22 {"technique": "semantic", "query": "What NumPy component had a string comparison bug and when was it fixed?", "rank": 1, "store_score": 0.64, "cosine": 0.9999999999999999}
23 {"technique": "semantic", "query": "What NumPy component had a string comparison bug and when was it fixed?", "rank": 2, "store_score": 0.35, "cosine": 0.9999999999999999}
24 {"technique": "semantic", "query": "What NumPy component had a string comparison bug and when was it fixed?", "rank": 3, "store_score": 0.35, "cosine": 0.9999999999999999}
25 {"technique": "semantic", "query": "Which package has the most vulnerabilities in the datasets?", "rank": 1, "store_score": 0.3858228579573216, "cosine": 0.9999999999999999}
26 {"technique": "semantic", "query": "Which package has the most vulnerabilities in the datasets?", "rank": 2, "store_score": 0.346891320471132, "cosine": 0.9999999999999999}
27 {"technique": "semantic", "query": "Which package has the most vulnerabilities in the datasets?", "rank": 3, "store_score": 0.337165732507738, "cosine": 0.9999999999999999}
28 {"technique": "semantic", "query": "Name two packages with RCE related vulnerabilities in SSH or networking code.", "rank": 1, "store_score": 0.64, "cosine": 0.9999999999999999}
29 {"technique": "semantic", "query": "Name two packages with RCE related vulnerabilities in SSH or networking code.", "rank": 2, "store_score": 0.64, "cosine": 0.9999999999999999}
30 {"technique": "semantic", "query": "Name two packages with RCE related vulnerabilities in SSH or networking code.", "rank": 3, "store_score": 0.64, "cosine": 0.9999999999999999}
31 {"technique": "sentenceWindow", "query": "What PyYAML method could allow arbitrary code execution and when was it fixed?", "rank": 1, "store_score": 0.7504473933474267, "cosine": 0.9999999999999999}
32 {"technique": "sentenceWindow", "query": "What PyYAML method could allow arbitrary code execution and when was it fixed?", "rank": 2, "store_score": 0.7504473933474267, "cosine": 0.9999999999999999}
33 {"technique": "sentenceWindow", "query": "What PyYAML method could allow arbitrary code execution and when was it fixed?", "rank": 3, "store_score": 0.7504473933474267, "cosine": 0.9999999999999999}
34 {"technique": "sentenceWindow", "query": "What bug affected Paramiko's write_private_key_file?", "rank": 1, "store_score": 0.789196487515887, "cosine": 0.9999999999999999}
35 {"technique": "sentenceWindow", "query": "What bug affected Paramiko's write_private_key_file?", "rank": 2, "store_score": 0.641273022260004, "cosine": 0.9999999999999999}
36 {"technique": "sentenceWindow", "query": "What bug affected Paramiko's write_private_key_file?", "rank": 3, "store_score": 0.62219877579132, "cosine": 0.9999999999999999}
37 {"technique": "sentenceWindow", "query": "What NumPy component had a string comparison bug and when was it fixed?", "rank": 1, "store_score": 0.64, "cosine": 0.9999999999999999}
38 {"technique": "sentenceWindow", "query": "What NumPy component had a string comparison bug and when was it fixed?", "rank": 2, "store_score": 0.35, "cosine": 0.9999999999999999}
39 {"technique": "sentenceWindow", "query": "What NumPy component had a string comparison bug and when was it fixed?", "rank": 3, "store_score": 0.35, "cosine": 0.9999999999999999}
40 {"technique": "sentenceWindow", "query": "Which package has the most vulnerabilities in the datasets?", "rank": 1, "store_score": 0.4269172360, "cosine": 0.9999999999999999}
41 {"technique": "sentenceWindow", "query": "Which package has the most vulnerabilities in the datasets?", "rank": 2, "store_score": 0.402015441, "cosine": 0.9999999999999999}
42 {"technique": "sentenceWindow", "query": "Which package has the most vulnerabilities in the datasets?", "rank": 3, "store_score": 0.35207112, "cosine": 0.9999999999999999}
43 {"technique": "sentenceWindow", "query": "Name two packages with RCE related vulnerabilities in SSH or networking code.", "rank": 1, "store_score": 0.64, "cosine": 0.9999999999999999}
44 {"technique": "sentenceWindow", "query": "Name two packages with RCE related vulnerabilities in SSH or networking code.", "rank": 2, "store_score": 0.64, "cosine": 0.9999999999999999}
45 {"technique": "sentenceWindow", "query": "Name two packages with RCE related vulnerabilities in SSH or networking code.", "rank": 3, "store_score": 0.64, "cosine": 0.9999999999999999}
46
data260-0932 > reports > hw03 > raw > {} per_technique_results.json > ...
1 {"token": {"num_chunks": 438, "avg_chunk_len": 662.6369863013699, "mean_latency_ms": 11.83930000042892}, "semantic": {"num_chunks": 118, "avg_chunk_len": 662.6369863013699, "mean_latency_ms": 11.83930000042892}}

```

Full output look for the json files are on the github

Query to use

Before running, create questions.yaml with five realistic domain questions (relevant to your DOMAIN_ID). For each question, record an expected answer and expected source file; at least two questions must depend on information found in only one source document. Commit questions.yaml before adding the retrieval results so the expected answers are not rewritten after seeing the output.

```

data260-0932 > reports > hw03 > ! questions.yaml
1 - id: q1
2   question: "What PyYAML method could allow arbitrary code execution and when was it fixed?"
3   expected_answer: "full_load / FullLoader; fixed in PyYAML 5.3.1"
4   expected_source_file: "dataset/PyYAML.txt"
5   single_source_only: true
6 - id: q2
7   question: "What bug affected Paramiko's write_private_key_file?"
8   expected_answer: "fA race condition between file creation and chmod that code cause information disclosure"
9   expected_source_file: "dataset/paramiko.txt"
10  single_source_only: true
11 - id: q3
12   question: "What NumPy component had a string comparision bug and when was it fixed?"
13   expected_answer: "numpy.core; fixed in NumPy 1.22"
14   expected_source_file: "dataset/numpy.txt"
15   single_source_only: true
16 - id: q4
17   question: "Which package has the most vulnerabilites in the datasets?"
18   expected_answer: "Pillow, which has 52 vulnerabilites"
19   expected_source_file: "dataset/pillow.txt"
20   single_source_only: false
21 - id: q5
22   question: "Name two packages with RCE related vulnerabilites in SSH or networking code."
23   expected_answer: "Paramiko and Twisted"
24   expected_source_file: "dataset/paramiko.txt and dataset/Twsited"
25   single_source_only: false

```

Output using the yaml file is the output screenshots above

What to compare (report section)

After you run the three pipelines:

1. Retrieval Quality:

- top-1 cosine (highest similarity among the top-k for that technique)
- mean@k cosine (average of top-k cosines)
- #chunks produced by the chunker and the avg chunk length (characters or tokens)
- retrieval latency in milliseconds (time the similarity search took; simple timer is fine)

```

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw03> python .\metrics.py
# METRICS.md

## Retrieval quality summary

| Technique | Chunks | Avg chunk length (chars) | Top-1 cosine | Mean@3 cosine | Recall@3 | Mean retrieval latency (ms) |
| ---|---|---|---|---|---|---|
| token | 438 | 663 | 0.620 | 0.528 | 0.000 | 11.84 |
| semantic | 118 | 1972 | 0.574 | 0.488 | 0.000 | 7.59 |
| sentenceWindow | 1031 | 226 | 0.619 | 0.475 | 0.000 | 21.04 |

```

```

1 # METRICS.md
2
3 ## Retrieval quality summary
4
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10

```

```

data260-0932 > reports > hw03 > metrics.py > main
1 import json
2 from collections import defaultdict
3 import yaml
4
5 def recall_at_k(technique, correct_answer, by_technique, k=3):
6     hits=0
7     total=0
8     for question, correct in correct_answer.items():
9         total+=1
10        retrieved = [
11            r["source_file"] for r in by_technique[technique]
12            if r["query"] == question and r["rank"] <=k
13        ]
14        if any(f in retrieved for f in correct):
15            hits+=1
16    if total == 0:
17        return 0.0
18    return hits/total
19
20 def main():
21     #getting correct answer for each query question
22     questions = yaml.safe_load(open("questions.yaml"))
23     correct_answer={}
24     for q in questions:
25         #if more than one correct answer for the file
26         files=[f.strip() for f in q["expected_source_file"].split(",")]
27         correct_answer[q["question"]] = files
28     with open("raw/per_query_results.json", encoding="utf-8") as f:
29         rows=[json.loads(line) for line in f if line.strip()]
30     with open("raw/per_technique_results.json", encoding="utf-8") as f:
31         technique=json.load(f)
32
33     # group cosine scores by technique
34     by_technique = defaultdict(list)
35     for r in rows:
36         by_technique[r["technique"]].append(r)
37
38     lines = []
39     lines.append("# METRICS.md\n")
40     lines.append("## Retrieval quality summary\n")
41     lines.append("| Technique | Chunks | Avg chunk length (chars) | Top-1 cosine | Mean@3 cosine | Recall@3 | Mean retrieval latency (m) |")
42     lines.append("|---|---|---|---|---|---|---|")
43
44     for name in ["token", "semantic", "sentenceWindow"]:
45         group = by_technique[name]
46         top1 = [r["cosine_sim"] for r in group if r["rank"] == 1]
47         all_cos = [r["cosine_sim"] for r in group]
48         stats = technique[name]
49         recall = recall_at_k(name, correct_answer, by_technique, k=3)
50         lines.append(f"| {name} | {stats['num_chunks']} | {stats['avg_chunk_len']:.0f} | {sum(top1)/len(top1):.3f} | {sum(all_cos)/len(all_cos):.3f} | {stats['mean_latency']:.2f} |")
51     )
52
53     open("METRICS.md", "w").write("\n".join(lines) + "\n")
54     print("\n".join(lines))
55 if __name__ == "__main__":
56     main()

```

Find at least one confidently scored retrieval that does not contain the answer. If your five questions don't produce one, add questions until you find one. Explain why the embedding likely considered it similar.

```

token | query: 'Which package has the most vulnerabilities in the datasets?'
query vector shape: (384,), first 8 values: [-0.051  0.006 -0.069 -0.05  0.05 -0.032 -0.009  0.08 ]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.386, 0.42865609164071883, 376, package: numpy (version:numpy/1.21.0) vulnerability count: 1 id: GHSA-fpfv-jqm9-f5jm aliases: CVE-2021-34141, PYSEC-2021-855 summary: None details: Incomplete
2, 0.366, 0.3765998807712292, 597, package: pillow (version:pillow/8.2.0) vulnerability count: 52 id: PYSEC-2021-331 aliases: CVE-2021-34552, GHSA-7534-mm45-c74v summary: None details: Pillow th
3, 0.356, 0.39722918619797513, 574, BIT-pillow-2022-45198, CVE-2022-45198, GHSA-m2vv-5vj5-2hm7 summary: None details: Pillow before 9.2.0 performs Improper Handling of Highly Compressed GIF Data (

```

For the query “Which package has the most vulnerabilities in the dataset?”, I noticed that the first rank labeled the package numpy as the answer. This is incorrect and the cosine similarity is 0.429 approximately. It shows that there is one vulnerability for NumPy, while the Pillow package has 52 vulnerabilities. The Pillow package answer is ranked below the NumPy one which shows that the most confident score retrieval does not contain the correct answer. Its most likely considered similar because the keywords in the query hit the package.txt file. The

embedding only accounts for the wording similarity, so the vulnerability count value is not used for the comparison. Therefore, NumPy was ranked first.

2. Observations (1–2 short paragraphs):

○ *Discuss why one technique performed better on this query (e.g., sentence coherence, semantic boundary detection, token-budget alignment, context carried via sentence window).*

```
token | query: "What bug affected Paramiko's write_private_key_file?"
query vector shape: (384,), first 8 values: [-0.04 -0.006 -0.085 0.032 0.032 -0.03 -0.02 0.095]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.750, 0.729018762146011, 530, package: paramiko (version:paramiko/2.4.1) vulnerability count: 4 id: PYSEC-2022-166 aliases: CVE-2022-24302, GHSA-f8q4-jwww-x3wv summary: None details: In Par
2, 0.567, 0.5682809410690168, 491, None details: Paramiko version 2.4.1, 2.3.2, 2.2.3, 2.1.5, 2.0.8, 1.18.5, 1.17.6 contains a Incorrect Access Control vulnerability in SSH server that can result
3, 0.545, 0.4699150282476977, 426, PYSEC-2026-2858 summary: None details: In Paramiko through 4.0.0 before a448945, rsakey.py allows the SHA-1 algorithm. vulnerability fixed in versions: links:
```

```
semantic | query: "What bug affected Paramiko's write_private_key_file?"
query vector shape: (384,), first 8 values: [-0.04 -0.006 -0.085 0.032 0.032 -0.03 -0.02 0.095]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.733, 0.7150091588376839, 704, package: paramiko (version:paramiko/2.4.1) vulnerability count: 4 id: PYSEC-2022-166 aliases: CVE-2022-24302, GHSA-f8q4-jwww-x3wv summary: None details: In Par
2, 0.575, 0.5331728596991491, 578, vulnerability fixed in versions: 2.4.2, 2.3.3, 2.2.4, 2.1.6, 2.0.9 links: ----- id: GHSA-r374-rxx8-8654 aliases: CVE-2026-44405, PYSEC-20
3, 0.430, 0.4013753060649975, 2676, This can lead to security issues in various situations, such as the most commonly used signature verification (ECDSA) and shared key negotiation (ECDH). When th
```

```
sentenceWindow | query: "What bug affected Paramiko's write_private_key_file?"
query vector shape: (384,), first 8 values: [-0.04 -0.006 -0.085 0.032 0.032 -0.03 -0.02 0.095]
stacked vectors shape: (3, 384)
rank, store_score, cosine_sim, chunk_len, preview
1, 0.789, 0.771990802247894, 315, package: paramiko (version:paramiko/2.4.1) vulnerability count: 4 id: PYSEC-2022-166 aliases: CVE-2022-24302, GHSA-f8q4-jwww-x3wv summary: None details: In Par
2, 0.641, 0.32539294028804276, 69, vulnerability fixed in versions: links: -----
3, 0.622, 0.5613412682848823, 271, vulnerability fixed in versions: 2.4.2, 2.3.3, 2.2.4, 2.1.6, 2.0.9 links: ----- id: GHSA-r374-rxx8-8654 aliases: CVE-2026-44405, PYSEC-20
```

The results of the pipeline show that the sentence window retrieval did the best when answering the query "What bug affected Paramiko's write_private_key_file?". Because the query is simple, I believe the cosine similarity for sentence window technique was ranked first with a value of 0.7712. Token and semantic respectively had 0.7290 and 0.7150. It appears the sentence window result showed 315 characters, which is considerably less compared to the other techniques. I believe with less characters it was able to find the answer faster with less of a chance of it getting unnecessary information.

○ *If the best technique differs across your optional extra queries, mention it.*

The best technique is the sentence window technique. By looking at all the results, I see that the top two cosine similarities are 0.7712 and 0.7256 for the sentence window technique. These cosine similarities are for the queries asking about Paramiko and PyYAML packages specifically. The query about NumPy and which packages have the most vulnerabilities show that the token technique is the best. These cosine similarities are the third and fourth highest. The values are 0.6878 and 0.4287. There is a noticeable gap between the top 2 and the third and fourth highest. Lastly, the last question about the RCE has a fifth highest cosine similarity value of 0.4515. The best technique for this question is the sentence window method.

3. Your conclusion (2–5 sentences):

○ *State which technique you judge best for this corpus and why supported by your measurements.*

For this whole experiment, I think the sentence window technique is the best because it showed the strongest cosine similarity out of all three techniques. Out of the five questions, sentence window technique had the highest cosine similarity of all three techniques for 3 out of 5 questions. Additionally, the top two cosine similarity values are from the sentence window technique. The values are 0.7712 and 0.7256 for the query questions about Paramiko and PyYAML packages specifically. I believed this technique is also the best because of the smaller

chunks it analyzed and its being able to keep surrounding context available was useful for finding the package answer to the questions more easily.

Verification

```
data260-0932 > reports > hw03 > verify-hw03.py > main
```

```
1 import json
2 import os
3
4 def main():
5
6     checks={}
7     checks["dataset_folder_exists"] = os.path.isdir("dataset")
8     checks["dataset_200kb_size"] = sum(os.path.getsize(os.path.join("dataset",f)) for f in os.listdir("dataset"))
9     checks["SOURCES_file_exists"] = os.path.exists("SOURCES.md")
10    checks["corpus_manifest_exists"] = os.path.exists("CORPUS_MANIFEST.json")
11    checks["questions_file_exists"] = os.path.exists("questions.yaml")
12    checks["metrics_file_exists"] = os.path.exists("METRICS.md")
13
14    checks["all_files_created_successfully"] = all(v for k, v in checks.items() if isinstance(v,bool))
15    json.dump(checks,open("verification.json", "w"))
16    print(json.dumps(checks))
17 if __name__=="__main__":
18     main()
```

```
data260-0932 > reports > hw03 > {} verification.json > ...
```

```
1 {"dataset_folder_exists": true, "dataset_200kb_size": true, "SOURCES_file_exists": true, "corpus_manifest_exists": true, "questions_file_exists": true, "metrics_file_exists": true, "all_files_created_successfully": true}
```

```
data260-0932 > reports > hw03 > {} verification.json > ...
```

```
1 true, "questions_file_exists": true, "metrics_file_exists": true, "all_files_created_successfully": true}
```